Neural Computation -- 2005

131 article(s), all issues

1. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-5

No abstract available.

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2. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-12

No abstract available.

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3. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-1

No abstract available.

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4. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-7

No abstract available.

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5. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-10

No abstract available.

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6. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-6

No abstract available.

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7. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-2

No abstract available.

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8. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-11

No abstract available.

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9. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-4

No abstract available.

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10. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-3

No abstract available.

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11. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-9

No abstract available.

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12. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2005.17.issue-8

No abstract available.

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13. Mixture Models Based on Neural Network Averaging

Walter W. Focke

https://doi.org/10.1162/089976606774841576 | pp. 1-9

A modified version of the single hidden-layer perceptron architecture is proposed for modeling mixtures. A particular flexible mixture model is obtained by implementing the Box-Cox transformation as transfer function. In this case, the network response can be expressed in closed form as a weighted power mean. The quadratic Scheffé K-polynomial and the exponential Wilson equation turn out to be special forms of this general mixture model. Advantages of the proposed network architecture are that binary data sets suffice for “training” and that it is readily extended to incorporate additional mixture components while retaining all previously determined weights.

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14. Sensory Memory for Odors Is Encoded in Spontaneous Correlated Activity Between Olfactory Glomeruli

Roberto F. Galán, Marcel Weidert, Randolf Menzel, Andreas V. M. Herz, C. Giovanni Galizia

https://doi.org/10.1162/089976606774841558 | pp. 10-25

Sensory memory is a short-lived persistence of a sensory stimulus in the nervous system, such as iconic memory in the visual system. However, little is known about the mechanisms underlying olfactory sensory memory. We have therefore analyzed the effect of odor stimuli on the first odor-processing network in the honeybee brain, the antennal lobe, which corresponds to the vertebrate olfactory bulb. We stained output neurons with a calcium-sensitive dye and measured across-glomerular patterns of spontaneous activity before and after a stimulus. Such a single-odor presentation changed the relative timing of spontaneous activity across glomeruli in accordance with Hebb's theory of learning. Moreover, during the first few minutes after odor presentation, correlations between the spontaneous activity fluctuations suffice to reconstruct the stimulus. As spontaneous activity is ubiquitous in the brain, modifiable fluctuations could provide an ideal substrate for Hebbian reverberations and sensory memory in other neural systems.

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15. The Optimal Synapse for Sparse, Binary Signals in the Rod Pathway

Paul T. Clark, Mark C. W. van Rossum

https://doi.org/10.1162/089976606774841530 | pp. 26-44

The sparsity of photons at very low light levels necessitates a nonlinear synaptic transfer function between the rod photoreceptors and the rod-bipolar cells. We examine different ways to characterize the performance of the pathway: the error rate, two variants of the mutual information, and the signal-to-noise ratio. Simulation of the pathway shows that these approaches yield substantially different performance at very low light levels and that maximizing the signal-to-noise ratio yields the best performance when judged from simulated images. The results are compared to recent data.

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16. Simultaneous Rate-Synchrony Codes in Populations of Spiking Neurons

Naoki Masuda

https://doi.org/10.1162/089976606774841521 | pp. 45-59

Firing rates and synchronous firing are often simultaneously relevant signals, and they independently or cooperatively represent external sensory inputs, cognitive events, and environmental situations such as body position. However, how rates and synchrony comodulate and which aspects of inputs are effectively

encoded, particularly in the presence of dynamical inputs, are unanswered questions. We examine theoretically how mixed information in dynamic mean input and noise input is represented by dynamic population firing rates and synchrony. In a subthreshold regime, amplitudes of spatially uncorrelated noise are encoded up to a fairly high input frequency, but this requires both rate and synchrony output channels. In a suprathreshold regime, means and common noise amplitudes can be simultaneously and separately encoded by rates and synchrony, respectively, but the input frequency for which this is possible has a lower limit.

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17. Spontaneous Dynamics of Asymmetric Random Recurrent Spiking Neural Networks

Hédi Soula, Guillaume Beslon, Olivier Mazet

https://doi.org/10.1162/089976606774841567 | pp. 60-79

In this letter, we study the effect of a unique initial stimulation on random recurrent networks of leaky integrate-and-fire neurons. Indeed, given a stochastic connectivity, this so-called spontaneous mode exhibits various nontrivial dynamics. This study is based on a mathematical formalism that allows us to examine the variability of the afterward dynamics according to the parameters of the weight distribution. Under the independence hypothesis (e.g., in the case of very large networks), we are able to compute the average number of neurons that fire at a given time-the spiking activity. In accordance with numerical simulations, we prove that this spiking activity reaches a steady state. We characterize this steady state and explore the transients.

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18. Bayesian Population Decoding of Motor Cortical Activity Using a Kalman Filter

Wei Wu, Yun Gao, Elie Bienenstock, John P. Donoghue, Michael J. Black

https://doi.org/10.1162/089976606774841585 | pp. 80-118

Effective neural motor prostheses require a method for decoding neural activity representing desired movement. In particular, the accurate reconstruction of a continuous motion signal is necessary for the control of devices such as computer cursors, robots, or a patient's own paralyzed limbs. For such applications, we developed a real-time system that uses Bayesian inference techniques to estimate hand motion from the firing rates of multiple neurons. In this study, we used recordings that were previously made in the arm area of primary motor cortex in awake behaving monkeys using a chronically implanted multielectrode microarray. Bayesian inference involves computing the posterior probability of the hand motion conditioned on a sequence of observed firing rates; this is formulated in terms of the product of a likelihood and a prior. The likelihood term models the probability of firing rates given a particular hand motion. We found that a linear gaussian model could be used to approximate this likelihood and could be readily learned from a small amount of training data. The prior term defines a probabilistic model of hand kinematics and was also taken to be a linear gaussian model. Decoding was performed using a Kalman filter, which gives an efficient recursive method for Bayesian inference when the likelihood and prior are linear and gaussian. In off-line experiments, the Kalman filter reconstructions of hand trajectory were more accurate than previously reported results. The resulting decoding algorithm provides a principled probabilistic model of motor-cortical coding, decodes hand motion in real time, provides an estimate of

uncertainty, and is straightforward to implement. Additionally the formulation unifies and extends previous models of neural coding while providing insights into the motor-cortical code.

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19. Facial Attractiveness: Beauty and the Machine

Yael Eisenthal, Gideon Dror, Eytan Ruppin

https://doi.org/10.1162/089976606774841602 | pp. 119-142

This work presents a novel study of the notion of facial attractiveness in a machine learning context. To this end, we collected human beauty ratings for data sets of facial images and used various techniques for learning the attractiveness of a face. The trained predictor achieves a significant correlation of 0.65 with the average human ratings. The results clearly show that facial beauty is a universal concept that a machine can learn. Analysis of the accuracy of the beauty prediction machine as a function of the size of the training data indicates that a machine producing human-like attractiveness rating could be obtained given a moderately larger data set.

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20. Classification of Faces in Man and Machine

Arnulf Graf, Felix A. Wichmann, HH Bülthoff, Bernhard Schölkopf

https://doi.org/10.1162/089976606774841611 | pp. 143-165

We attempt to shed light on the algorithms humans use to classify images of human faces according to their gender. For this, a novel methodology combining human psychophysics and machine learning is introduced. We proceed as follows. First, we apply principal component analysis (PCA) on the pixel information of the face stimuli. We then obtain a data set composed of these PCA eigenvectors combined with the subjects' gender estimates of the corresponding stimuli. Second, we model the gender classification process on this data set using a separating hyperplane (SH) between both classes. This SH is computed using algorithms from machine learning: the support vector machine (SVM), the relevance vector machine, the prototype classifier, and the K-means classifier. The classification behavior of humans and machines is then analyzed in three steps. First, the classification errors of humans and machines are compared for the various classifiers, and we also assess how well machines can recreate the subjects' internal decision boundary by studying the training errors of the machines. Second, we study the correlations between the rank-order of the subjects' responses to each stimulus-the gender estimate with its reaction time and confidence rating-and the rank-order of the distance of these stimuli to the SH. Finally, we attempt to compare the metric of the representations used by humans and machines for classification by relating the subjects' gender estimate of each stimulus and the distance of this stimulus to the SH. While we show that the classification error alone is not a sufficient selection criterion between the different algorithms humans might use to classify face stimuli, the distance of these stimuli to the SH is shown to capture essentials of the internal decision space of humans. Furthermore, algorithms such as the prototype classifier using stimuli in the center of the classes are shown to be less adapted to model human classification behavior than algorithms such as the SVM based on stimuli close to the boundary between the classes.

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21. Exploring Latent Structure of Mixture ICA Models by the Minimum β-Divergence Method

Md. Nurul Haque Mollah, Mihoko Minami, Shinto Eguchi

https://doi.org/10.1162/089976606774841549 | pp. 166-190

Independent component analysis (ICA) attempts to extract original independent signals (source components) that are linearly mixed in a basic framework. This letter discusses a learning algorithm for the separation of different source classes in which the observed data follow a mixture of several ICA models, where each model is described by a linear combination of independent and nongaussian sources. The proposed method is based on a sequential application of the minimum β-divergence method to separate all source classes sequentially. The proposed method searches the recovering matrix of each class on the basis of a rule of sequential change of the shifting parameter. If the initial choice of the shifting parameter vector is close to the mean of a data class, then all of the hidden sources belonging to that class are recovered properly with independent and nongaussian structure considering the data in other classes as outliers. The value of the tuning parameter β is a key in the performance of the proposed method. A cross-validation technique is proposed as an adaptive selection procedure for the tuning parameter β for this algorithm, together with applications for both real and synthetic data analysis.

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22. An Adaptive Method for Subband Decomposition ICA

Kun Zhang, Laiwan Chan

https://doi.org/10.1162/089976606774841620 | pp. 191-223

Subband decomposition ICA (SDICA), an extension of ICA, assumes that each source is represented as the sum of some independent subcomponents and dependent subcomponents, which have different frequency bands. In this article, we first investigate the feasibility of separating the SDICA mixture in an adaptive manner. Second, we develop an adaptive method for SDICA, namely band-selective ICA (BS-ICA), which finds the mixing matrix and the estimate of the source independent subcomponents. This method is based on the minimization of the mutual information between outputs. Some practical issues are discussed. For better applicability, a scheme to avoid the high-dimensional score function difference is given. Third, we investigate one form of the overcomplete ICA problems with sources having specific frequency characteristics, which BS-ICA can also be used to solve. Experimental results illustrate the success of the proposed method for solving both SDICA and the over-complete ICA problems.

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23. On Consistency of Bayesian Inference with Mixtures of Logistic Regression

Yang Ge, Wenxin Jiang

https://doi.org/10.1162/089976606774841594 | pp. 224-243

This is a theoretical study of the consistency properties of Bayesian inference using mixtures of logistic regression models. When standard logistic regression models are combined in a mixtures-of-experts setup, a flexible model is formed to model the relationship between a binary (yes-no) response y and a vector of predictors x. Bayesian inference conditional on the observed data can then be used for regression and classification. This letter gives conditions on choosing the number of experts (i.e., number of mixing

components) k or choosing a prior distribution for k, so that Bayesian inference is consistent, in the sense of often approximating the underlying true relationship between y and x. The resulting classification rule is also consistent, in the sense of having near-optimal performance in classification. We show these desirable consistency properties with a nonstochastic k growing slowly with the sample size n of the observed data, or with a random k that takes large values with nonzero but small probabilities.

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24. Polychronization: Computation with Spikes

Eugene M. Izhikevich

https://doi.org/10.1162/089976606775093882 | pp. 245-282

We present a minimal spiking network that can polychronize, that is, exhibit reproducible time-locked but not synchronous firing patterns with millisecond precision, as in synfire braids. The network consists of cortical spiking neurons with axonal conduction delays and spike-timing-dependent plasticity (STDP); a ready-to-use MATLAB code is included. It exhibits sleeplike oscillations, gamma (40 Hz) rhythms, conversion of firing rates to spike timings, and other interesting regimes. Due to the interplay between the delays and STDP, the spiking neurons spontaneously self-organize into groups and generate patterns of stereotypical polychronous activity. To our surprise, the number of coexisting polychronous groups far exceeds the number of neurons in the network, resulting in an unprecedented memory capacity of the system. We speculate on the significance of polychrony to the theory of neuronal group selection (TNGS, neural Darwinism), cognitive neural computations, binding and gamma rhythm, mechanisms of attention, and consciousness as "attention to memories."

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25. Temporal Sequence Learning, Prediction, and Control: A Review of Different Models and Their Relation to Biological Mechanisms

Florentin Wörgötter, Bernd Porr

https://doi.org/10.1162/0899766053011555 | pp. 245-319

In this review, we compare methods for temporal sequence learning (TSL) across the disciplines machine-control, classical conditioning, neuronal models for TSL as well as spike-timing-dependent plasticity (STDP). This review introduces the most influential models and focuses on two questions: To what degree are reward-based (e.g., TD learning) and correlation-based (Hebbian) learning related? and How do the different models correspond to possibly underlying biological mechanisms of synaptic plasticity? We first compare the different models in an open-loop condition, where behavioral feedback does not alter the learning. Here we observe that reward-based and correlation-based learning are indeed very similar. Machine control is then used to introduce the problem of closed-loop control (e.g., actor-critic architectures). Here the problem of evaluative (rewards) versus nonevaluative (correlations) feedback from the environment will be discussed, showing that both learning approaches are fundamentally different in the closed-loop condition. In trying to answer the second question, we compare neuronal versions of the different learning architectures to the anatomy of the involved brain structures (basal-ganglia, thalamus, and cortex) and the molecular biophysics of glutamatergic and dopaminergic synapses. Finally, we discuss the different algorithms used to model STDP and compare them to reward-based learning rules. Certain similarities are found in spite of the strongly different timescales. Here we focus on the biophysics of the different calcium-release mechanisms known to be involved in STDP.

26. Making Working Memory Work: A Computational Model of Learning in the Prefrontal Cortex and Basal Ganglia

Randall C. O'Reilly, Michael J. Frank

https://doi.org/10.1162/089976606775093909 | pp. 283-328

The prefrontal cortex has long been thought to subserve both working memory (the holding of information online for processing) and executive functions (deciding how to manipulate working memory and perform processing). Although many computational models of working memory have been developed, the mechanistic basis of executive function remains elusive, often amounting to a homunculus. This article presents an attempt to deconstruct this homunculus through powerful learning mechanisms that allow a computational model of the prefrontal cortex to control both itself and other brain areas in a strategic, task-appropriate manner. These learning mechanisms are based on subcortical structures in the midbrain, basal ganglia, and amygdala, which together form an actor-critic architecture. The critic system learns which prefrontal representations are task relevant and trains the actor, which in turn provides a dynamic gating mechanism for controlling working memory updating. Computationally, the learning mechanism is designed to simultaneously solve the temporal and structural credit assignment problems. The model's performance compares favorably with standard backpropagation-based temporal learning mechanisms on the challenging 1-2-AX working memory task and other benchmark working memory tasks.

27. A Note on Stone's Conjecture of Blind Signal Separation

Shengli Xie, Zhaoshui He, Yuli Fu

https://doi.org/10.1162/0899766053011492 | pp. 321-330

Stone's method is one of the novel approaches to the blind source separation (BSS) problem and is based on Stone's conjecture. However, this conjecture has not been proved. We present a simple simulation to demonstrate that Stone's conjecture is incorrect. We then modify Stone's conjecture and prove this modified conjecture as a theorem, which can be used a basis for BSS algorithms.

28. Identification of Multiple-Input Systems with Highly Coupled Inputs: Application to EMG Prediction from Multiple Intracortical Electrodes

David T. Westwick, Eric A. Pohlmeyer, Sara A. Solla, Lee E. Miller, Eric J. Perreault

https://doi.org/10.1162/089976606775093855 | pp. 329-355

A robust identification algorithm has been developed for linear, time-invariant, multiple-input single-output systems, with an emphasis on how this algorithm can be used to estimate the dynamic relationship between a set of neural recordings and related physiological signals. The identification algorithm provides a decomposition of the system output such that each component is uniquely attributable to a specific input signal, and then reduces the complexity of the estimation problem by discarding those input signals that are deemed to be insignificant. Numerical difficulties due to limited input bandwidth and correlations among the inputs are addressed using a robust estimation technique based on singular value decomposition. The algorithm has been evaluated on both simulated and experimental data. The latter involved estimating the relationship between up to 40 simultaneously recorded motor cortical signals and peripheral

electromyograms (EMGs) from four upper limb muscles in a freely moving primate. The algorithm performed well in both cases: it provided reliable estimates of the system output and significantly reduced the number of inputs needed for output prediction. For example, although physiological recordings from up to 40 different neuronal signals were available, the input selection algorithm reduced this to 10 neuronal signals that made significant contributions to the recorded EMGs.

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29. A Further Result on the ICA One-Bit-Matching Conjecture

Jinwen Ma, Zhiyong Liu, Lei Xu

https://doi.org/10.1162/0899766053011483 | pp. 331-334

The one-bit-matching conjecture for independent component analysis (ICA) has been widely believed in the ICA community. Theoretically, it has been proved that under the assumption of zero skewness for the model probability density functions, the global maximum of a cost function derived from the typical objective function on the ICA problem with the one-bit-matching condition corresponds to a feasible solution of the ICA problem. In this note, we further prove that all the local maximums of the cost function correspond to the feasible solutions of the ICA problem in the two-source case under the same assumption. That is, as long as the one-bit-matching condition is satisfied, the two-source ICA problem can be successfully solved using any local descent algorithm of the typical objective function with the assumption of zero skewness for all the model probability density functions.

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30. Robust Reinforcement Learning

Jun Morimoto, Kenji Doya

https://doi.org/10.1162/0899766053011528 | pp. 335-359

This letter proposes a new reinforcement learning (RL) paradigm that explicitly takes into account input disturbance as well as modeling errors. The use of environmental models in RL is quite popular for both offline learning using simulations and for online action planning. However, the difference between the model and the real environment can lead to unpredictable, and often unwanted, results. Based on the theory of H(infinity) control, we consider a differential game in which a "disturbing" agent tries to make the worst possible disturbance while a "control" agent tries to make the best control input. The problem is formulated as finding a min-max solution of a value function that takes into account the amount of the reward and the norm of the disturbance. We derive online learning algorithms for estimating the value function and for calculating the worst disturbance and the best control in reference to the value function. We tested the paradigm, which we call robust reinforcement learning (RRL), on the control task of an inverted pendulum. In the linear domain, the policy and the value function learned by online algorithms coincided with those derived analytically by the linear H(infinity) control theory. For a fully nonlinear swing-up task, RRL achieved robust performance with changes in the pendulum weight and friction, while a standard reinforcement learning algorithm could not deal with these changes. We also applied RRL to the cart-pole swing-up task, and a robust swing-up policy was acquired.

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31. Oscillatory Networks: Pattern Recognition Without a Superposition Catastrophe

Thomas Burwick

https://doi.org/10.1162/089976606775093864 | pp. 356-380

Using an oscillatory network model that combines classical network models with phase dynamics, we demonstrate how the superposition catastrophe of pattern recognition may be avoided in the context of phase models. The model is designed to meet two requirements: on and off states should correspond, respectively, to high and low phase velocities, and patterns should be retrieved in coherent mode. Nonoverlapping patterns can be simultaneously active with mutually different phases. For overlapping patterns, competition can be used to reduce coherence to a subset of patterns. The model thereby solves the superposition problem.

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32. A Computational Model of the Functional Role of the Ventral-Striatal D2 Receptor in the Expression of Previously Acquired Behaviors

A. J. Smith, Suzanna Becker, Shitij Kapur

https://doi.org/10.1162/0899766053011546 | pp. 361-395

The functional role of dopamine has attracted a great deal of interest ever since it was empirically discovered that dopamine-blocking drugs could be used to treat psychosis. Specifically, the D2 receptor and its expression in the ventral striatum have emerged as pivotal in our understanding of the complex role of the neuromodulator in schizophrenia, reward, and motivation. Our departure from the ubiquitous temporal difference (TD) model of dopamine neuron firing allows us to account for a range of experimental evidence suggesting that ventral striatal dopamine D2 receptor manipulation selectively modulates motivated behavior for distal versus proximal outcomes. Whether an internal model or the TD approach (or a mixture) is better suited to a comprehensive exposition of tonic and phasic dopamine will have important implications for our understanding of reward, motivation, schizophrenia, and impulsivity. We also use the model to help unite some of the leading cognitive hypotheses of dopamine function under a computational umbrella. We have used the model ourselves to stimulate and focus new rounds of experimental research.

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33. Topographic Product Models Applied to Natural Scene Statistics

Simon Osindero, Max Welling, Geoffrey E. Hinton

https://doi.org/10.1162/089976606775093936 | pp. 381-414

We present an energy-based model that uses a product of generalized Student-t distributions to capture the statistical structure in data sets. This model is inspired by and particularly applicable to "natural" data sets such as images. We begin by providing the mathematical framework, where we discuss complete and overcomplete models and provide algorithms for training these models from data. Using patches of natural scenes, we demonstrate that our approach represents a viable alternative to independent component analysis as an interpretive model of biological visual systems. Although the two approaches are similar in flavor, there are also important differences, particularly when the representations are overcomplete. By constraining the interactions within our model, we are also able to study the topographic organization of Gabor-like receptive fields that our model learns. Finally, we discuss the relation of our new approach to

previous work--in particular, gaussian scale mixture models and variants of independent components analysis.

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34. A Hierarchical Bayesian Model for Learning Nonlinear Statistical Regularities in Nonstationary Natural Signals

Yan Karklin, Michael S. Lewicki

https://doi.org/10.1162/0899766053011474 | pp. 397-423

Capturing statistical regularities in complex, high-dimensional data is an important problem in machine learning and signal processing. Models such as principal component analysis (PCA) and independent component analysis (ICA) make few assumptions about the structure in the data and have good scaling properties, but they are limited to representing linear statistical regularities and assume that the distribution of the data is stationary. For many natural, complex signals, the latent variables often exhibit residual dependencies as well as nonstationary statistics. Here we present a hierarchical Bayesian model that is able to capture higher-order nonlinear structure and represent nonstationary data distributions. The model is a generalization of ICA in which the basis function coefficients are no longer assumed to be independent; instead, the dependencies in their magnitudes are captured by a set of density components. Each density component describes a common pattern of deviation from the marginal density of the pattern ensemble; in different combinations, they can describe nonstationary distributions. Adapting the model to image or audio data yields a nonlinear, distributed code for higher-order statistical regularities that reflect more abstract, invariant properties of the signal.

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35. A Simple Hebbian/Anti-Hebbian Network Learns the Sparse, Independent Components of Natural Images

Michael Falconbridge, R. L. Stamps, David R. Badcock

https://doi.org/10.1162/089976606775093891 | pp. 415-429

Slightly modified versions of an early Hebbian/anti-Hebbian neural network are shown to be capable of extracting the sparse, independent linear components of a prefiltered natural image set. An explanation for this capability in terms of a coupling between two hypothetical networks is presented. The simple networks presented here provide alternative, biologically plausible mechanisms for sparse, factorial coding in early primate vision.

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36. Extended Gaussianization Method for Blind Separation of Post-Nonlinear Mixtures

Kun Zhang, Laiwan Chan

https://doi.org/10.1162/0899766053011500 | pp. 425-452

The linear mixture model has been investigated in most articles tackling the problem of blind source separation. Recently, several articles have addressed a more complex model: blind source separation (BSS) of post-nonlinear (PNL) mixtures. These mixtures are assumed to be generated by applying an unknown invertible nonlinear distortion to linear instantaneous mixtures of some independent sources. The

gaussianization technique for BSS of PNL mixtures emerged based on the assumption that the distribution of the linear mixture of independent sources is gaussian. In this letter, we review the gaussianization method and then extend it to apply to PNL mixture in which the linear mixture is close to gaussian. Our proposed method approximates the linear mixture using the Cornish-Fisher expansion. We choose the mutual information as the independence measurement to develop a learning algorithm to separate PNL mixtures. This method provides better applicability and accuracy. We then discuss the sufficient condition for the method to be valid. The characteristics of the nonlinearity do not affect the performance of this method. With only a few parameters to tune, our algorithm has a comparatively low computation. Finally, we present experiments to illustrate the efficiency of our method.

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37. Differential Log Likelihood for Evaluating and Learning Gaussian Mixtures

Marc M. Van Hulle

https://doi.org/10.1162/089976606775093873 | pp. 430-445

We introduce a new unbiased metric for assessing the quality of density estimation based on gaussian mixtures, called differential log likelihood. As an application, we determine the optimal smoothness and the optimal number of kernels in gaussian mixtures. Furthermore, we suggest a learning strategy for gaussian mixture density estimation and compare its performance with log likelihood maximization for a wide range of real-world data sets.

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38. Magnification Control in Self-Organizing Maps and Neural Gas

Thomas Villmann, Jens Christian Claussen

https://doi.org/10.1162/089976606775093918 | pp. 446-469

We consider different ways to control the magnification in self-organizing maps (SOM) and neural gas (NG). Starting from early approaches of magnification control in vector quantization, we then concentrate on different approaches for SOM and NG. We show that three structurally similar approaches can be applied to both algorithms that are localized learning, concave-convex learning, and winner-relaxing learning. Thereby, the approach of concave-convex learning in SOM is extended to a more general description, whereas the concave-convex learning for NG is new. In general, the control mechanisms generate only slightly different behavior comparing both neural algorithms. However, we emphasize that the NG results are valid for any data dimension, whereas in the SOM case, the results hold only for the one-dimensional case.

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39. Bayesian Analysis of Nonlinear Autoregression Models Based on Neural Networks

Antonio Menchero, Raquel Montes Díez, David R■■os Insua, P. Müller

https://doi.org/10.1162/0899766053011537 | pp. 453-485

We show how Bayesian neural networks can be used for time-series analysis. We consider a block-based model building strategy to model linear and nonlinear features within the time series: a linear combination of a linear autoregression term and a feedforward neural network (FFNN) with an unknown number of hidden

nodes. To allow for simpler models, we also consider these terms separately as competing models to select from. Model identifiability problems arise when FFNN sigmoidal activation functions exhibit almost linear behavior or when there are almost duplicate or irrelevant neural network nodes. New reversible-jump moves are proposed to facilitate model selection, mitigating model identifiability problems. We illustrate this methodology analyzing several time-series data examples.

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40. Enhancing Density-Based Data Reduction Using Entropy

Di Huang, Tommy W. S. Chow

https://doi.org/10.1162/089976606775093927 | pp. 470-495

Data reduction algorithms determine a small data subset from a given large data set. In this article, new types of data reduction criteria, based on the concept of entropy, are first presented. These criteria can evaluate the data reduction performance in a sophisticated and comprehensive way. As a result, new data reduction procedures are developed. Using the newly introduced criteria, the proposed data reduction scheme is shown to be efficient and effective. In addition, an outlier-filtering strategy, which is computationally insignificant, is developed. In some instances, this strategy can substantially improve the performance of supervised data analysis. The proposed procedures are compared with related techniques in two types of application: density estimation and classification. Extensive comparative results are included to corroborate the contributions of the proposed algorithms.

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41. Loading Deep Networks Is Hard: The Pyramidal Case

David Windisch

https://doi.org/10.1162/0899766053011519 | pp. 487-502

The question of whether it is possible to load deep neural network architectures efficiently is examined by considering the class of pyramidal architectures. This class allows only a low interaction of the nodes. Still, the loading problem is found to be NP-complete. This provides evidence that depth alone is a factor accounting for loading hardness.

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42. Maximum Likelihood Topographic Map Formation

Marc M. Van Hulle

https://doi.org/10.1162/0899766053019935 | pp. 503-513

We introduce a new unsupervised learning algorithm for kernel-based topographic map formation of heteroscedastic gaussian mixtures that allows for a unified account of distortion error (vector quantization), log-likelihood, and Kullback-Leibler divergence.

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43. On Convergence Conditions of an Extended Projection Neural Network

Youshen Xia, Gang Feng

https://doi.org/10.1162/0899766053019926 | pp. 515-525

The output trajectory convergence of an extended projection neural network was developed under the positive definiteness condition of the Jacobian matrix of nonlinear mapping. This note offers several new convergence results. The state trajectory convergence and the output trajectory convergence of the extended projection neural network are obtained under the positive semidefiniteness condition of the Jacobian matrix. Comparison and illustrative examples demonstrate applied significance of these new results.

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44. Memorization and Association on a Realistic Neural Model

Leslie G. Valiant

https://doi.org/10.1162/0899766053019890 | pp. 527-555

A central open question of computational neuroscience is to identify the data structures and algorithms that are used in mammalian cortex to support successive acts of the basic cognitive tasks of memorization and association. This letter addresses the simultaneous challenges of realizing these two distinct tasks with the same data structure, and doing so while respecting the following four basic quantitative parameters of cortex: the neuron number, the synapse number, the synapse strengths, and the switching times. Previous work has not succeeded in reconciling these opposing constraints, the low values of synapse strengths that are typically observed experimentally having contributed a particular obstacle. In this article, we describe a computational scheme that supports both memory formation and association and is feasible on networks of model neurons that respect the widely observed values of the four quantitative parameters. Our scheme allows for both disjoint and shared representations. The algorithms are simple, and in one version both memorization and association require just one step of vicinal or neighborly influence. The issues of interference among the different circuits that are established, of robustness to noise, and of the stability of the hierarchical memorization process are addressed. A calculus therefore is implied for analyzing the capabilities of particular neural systems and subsystems, in terms of their basic numerical parameters.

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45. Effects of Noisy Drive on Rhythms in Networks of Excitatory and Inhibitory Neurons

Christoph Börgers, Nancy Kopell

https://doi.org/10.1162/0899766053019908 | pp. 557-608

Synchronous rhythmic spiking in neuronal networks can be brought about by the interaction between E-cells and Icells (excitatory and inhibitory cells). The I-cells gate and synchronize the E-cells, and the E-cells drive and synchronize the I-cells. We refer to rhythms generated in this way as PING (pyramidal-interneuronal gamma) rhythms. The PING mechanism requires that the drive I(I) to the I-cells be sufficiently low; the rhythm is lost when I(I) gets too large. This can happen in at least two ways. In the first mechanism, the I-cells spike in synchrony, but get ahead of the E-cells, spiking without being prompted by the E-cells. We call this phase walkthrough of the I-cells. In the second mechanism, the I-cells fail to synchronize, and their activity leads to complete suppression of the E-cells. Noisy spiking in the E-cells, generated by noisy external drive, adds excitatory drive to the I-cells and may lead to phase walkthrough. Noisy spiking in the I-cells adds inhibition to the E-cells and may lead to suppression of the E-cells. An analysis of the conditions under which noise leads to phase walkthrough of the I-cells or suppression of the E-cells shows that PING rhythms at frequencies far below the gamma range are robust to noise only if network parameter values are

tuned very carefully. Together with an argument explaining why the PING mechanism does not work far above the gamma range in the presence of heterogeneity, this justifies the "G" in "PING."

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46. Supervised Learning Through Neuronal Response Modulation

Christian D. Swinehart, L. F. Abbott

https://doi.org/10.1162/0899766053019980 | pp. 609-631

Neural networks that are trained to perform specific tasks must be developed through a supervised learning procedure. This normally takes the form of direct supervision of synaptic plasticity. We explore the idea that supervision takes place instead through the modulation of neuronal excitability. Such supervision can be done using conventional synaptic feedback pathways rather than requiring the hypothetical actions of unknown modulatory agents. During task learning, supervised response modulation guides Hebbian synaptic plasticity indirectly by establishing appropriate patterns of correlated network activity. This results in robust learning of function approximation tasks evenwhenmultiple output units representing different functions share large amounts of common input. Reward-based supervision is also studied, and a number of potential advantages of neuronal response modulation are identified.

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47. The Combined Effects of Inhibitory and Electrical Synapses in Synchrony

Benjamin Pfeuty, Germán Mato, David Golomb, D. Hansel

https://doi.org/10.1162/0899766053019917 | pp. 633-670

Recent experimental results have shown that GABAergic interneurons in the central nervous system are frequently connected via electrical synapses. Hence, depending on the area or the subpopulation, interneurons interact via inhibitory synapses or electrical synapses alone or via both types of interactions. The theoretical work presented here addresses the significance of these different modes of interactions for the interneuron networks dynamics. We consider the simplest system in which this issue can be investigated in models or in experiments: a pair of neurons, interacting via electrical synapses, inhibitory synapses, or both, and activated by the injection of a noisy external current. Assuming that the couplings and the noise are weak, we derive an analytical expression relating the cross-correlation (CC) of the activity of the two neurons to the phase response function of the neurons. When electrical and inhibitory interactions are not too strong, they combine their effect in a linear manner. In this regime, the effect of electrical and inhibitory interactions when combined can be deduced knowing the effects of each of the interactions separately. As a consequence, depending on intrinsic neuronal properties, electrical and inhibitory synapses may cooperate, both promoting synchrony, or may compete, with one promoting synchrony while the other impedes it. In contrast, for sufficiently strong couplings, the two types of synapses combine in a nonlinear fashion. Remarkably, we find that in this regime, combining electrical synapses with inhibition amplifies synchrony, whereas electrical synapses alone would desynchronize the activity of the neurons. We apply our theory to predict how the shape of the CC of two neurons changes as a function of ionic channel conductances, focusing on the effect of persistent sodium conductance, of the firing rate of the neurons and the nature and the strength of their interactions. These predictions may be tested using dynamic clamp techniques.

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48. Spikernels: Predicting Arm Movements by Embedding Population Spike Rate Patterns in Inner-Product Spaces

Lavi Shpigelman, Yoram Singer, Rony Paz, Eilon Vaadia

https://doi.org/10.1162/0899766053019944 | pp. 671-690

Inner-product operators, often referred to as kernels in statistical learning, define a mapping from some input space into a feature space. The focus of this letter is the construction of biologically motivated kernels for cortical activities. The kernels we derive, termed Spikernels, map spike count sequences into an abstract vector space in which we can perform various prediction tasks. We discuss in detail the derivation of Spikernels and describe an efficient algorithm for computing their value on any two sequences of neural population spike counts. We demonstrate the merits of our modeling approach by comparing the Spikernel to various standard kernels in the task of predicting hand movement velocities from cortical recordings. All of the kernels that we tested in our experiments outperform the standard scalar product used in linear regression, with the Spikernel consistently achieving the best performance.

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49. Memory Capacity of Balanced Networks

Yuval Aviel, D. Horn, Moshe Abeles

https://doi.org/10.1162/0899766053019962 | pp. 691-713

We study the problem of memory capacity in balanced networks of spiking neurons. Associative memories are represented by either synfire chains (SFC) or Hebbian cell assemblies (HCA). Both can be embedded in these balanced networks by a proper choice of the architecture of the network. The size w(E) of a pool in an SFC or of an HCA is limited from below and from above by dynamical considerations. Proper scaling of w(E) by radicalK, where K is the total excitatory synaptic connectivity, allows us to obtain a uniform description of our system for any given K. Using combinatorial arguments, we derive an upper limit on memory capacity. The capacity allowed by the dynamics of the system, alpha(c), is measured by simulations. For HCA, we obtain alpha(c) of order 0.1, and for SFC, we find values of order 0.065. The capacity can be improved by introducing shadow patterns, inhibitory cell assemblies that are fed by the excitatory assemblies in both memory models. This leads to a doubly balanced network, where, in addition to the usual global balancing of excitation and inhibition, there exists specific balance between the effects of both types of assemblies on the background activity of the network. For each of the memory models and for each network architecture, we obtain an allowed region (phase space) for w(E)/ radicalK in which the model is viable.

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50. On the Capabilities of Higher-Order Neurons: A Radial Basis Function Approach

Michael Schmitt

https://doi.org/10.1162/0899766053019953 | pp. 715-729

Higher-order neurons with k monomials in n variables are shown to have Vapnik-Chervonenkis (VC) dimension at least nk + 1. This result supersedes the previously known lower bound obtained via k-term monotone disjunctive normal form (DNF) formulas. Moreover, it implies that the VC dimension of higher-order neurons with k monomials is strictly larger than the VC dimension of k-term monotone DNF. The result is achieved by introducing an exponential approach that employs gaussian radial basis function

neural networks for obtaining classifications of points in terms of higher-order neurons.

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51. Estimating the Posterior Probabilities Using the K-Nearest Neighbor Rule

Amir F. Atiya

https://doi.org/10.1162/0899766053019971 | pp. 731-740

In many pattern classification problems, an estimate of the posterior probabilities (rather than only a classification) is required. This is usually the case when some confidence measure in the classification is needed. In this article, we propose a new posterior probability estimator. The proposed estimator considers the K-nearest neighbors. It attaches a weight to each neighbor that contributes in an additive fashion to the posterior probability estimate. The weights corresponding to the K-nearest-neighbors (which add to 1) are estimated from the data using a maximum likelihood approach. Simulation studies confirm the effectiveness of the proposed estimator.

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52. Quantifying Stimulus Discriminability: A Comparison of Information Theory and Ideal Observer Analysis

Eric E. Thomson, William B. Kristan

https://doi.org/10.1162/0899766053429435 | pp. 741-778

Performance in sensory discrimination tasks is commonly quantified using either information theory or ideal observer analysis. These two quantitative frameworks are often assumed to be equivalent. For example, higher mutual information is said to correspond to improved performance of an ideal observer in a stimulus estimation task. To the contrary, drawing on and extending previous results, we show that five information-theoretic quantities (entropy, response-conditional entropy, specific information, equivocation, and mutual information) violate this assumption. More positively, we show how these information measures can be used to calculate upper and lower bounds on ideal observer performance, and vice versa. The results show that the mathematical resources of ideal observer analysis are preferable to information theory for evaluating performance in a stimulus discrimination task. We also discuss the applicability of information theory to questions that ideal observer analysis cannot address.

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53. Nonlinear Complex-Valued Extensions of Hebbian Learning: An Essay

Simone Fiori

https://doi.org/10.1162/0899766053429381 | pp. 779-838

The Hebbian paradigm is perhaps the best-known unsupervised learning theory in connectionism. It has inspired wide research activity in the artificial neural network field because it embodies some interesting properties such as locality and the capability of being applicable to the basic weight-and-sum structure of neuron models. The plain Hebbian principle, however, also presents some inherent theoretical limitations that make it impractical in most cases. Therefore, modifications of the basic Hebbian learning paradigm have been proposed over the past 20 years in order to design profitable signal and data processing algorithms. Such modifications led to the principal component analysis type class of learning rules along with

their nonlinear extensions. The aim of this review is primarily to present part of the existing fragmented material in the field of principal component learning within a unified view and contextually to motivate and present extensions of previous works on Hebbian learning to complex-weighted linear neural networks. This work benefits from previous studies on linear signal decomposition by artificial neural networks, nonquadratic component optimization and reconstruction error definition, neural parameters adaptation by constrained optimization of learning criteria of complex-valued arguments, and orthonormality expression via the insertion of topological elements in the networks or by modifying the network learning criterion. In particular, the learning principles considered here and their analysis concern complex-valued principal/minor component/subspace linear/nonlinear rules for complex-weighted neural structures, both feedforward and laterally connected.

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54. Difficulty of Singularity in Population Coding

■■■-■■■ ■■■■■, *Hiroyuki Nakahara*

https://doi.org/10.1162/0899766053429426 | pp. 839-858

Fisher information has been used to analyze the accuracy of neural population coding. This works well when the Fisher information does not degenerate, but when two stimuli are presented to a population of neurons, a singular structure emerges by their mutual interactions. In this case, the Fisher information matrix degenerates, and the regularity condition ensuring the Cramér-Rao paradigm of statistics is violated. An animal shows pathological behavior in such a situation. We present a novel method of statistical analysis to understand information in population coding in which algebraic singularity plays a major role. The method elucidates the nature of the pathological case by calculating the Fisher information. We then suggest that synchronous firing can resolve singularity and show a method of analyzing the binding problem in terms of the Fisher information. Our method integrates a variety of disciplines in population coding, such as nonregular statistics, Bayesian statistics, singularity in algebraic geometry, and synchronous firing, under the theme of Fisher information.

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55. Neurons Tune to the Earliest Spikes Through STDP

Rudy Guyonneau, Rufin VanRullen, Simon J. Thorpe

https://doi.org/10.1162/0899766053429390 | pp. 859-879

Spike timing-dependent plasticity (STDP) is a learning rule that modifies the strength of a neuron's synapses as a function of the precise temporal relations between input and output spikes. In many brains areas, temporal aspects of spike trains have been found to be highly reproducible. How will STDP affect a neuron's behavior when it is repeatedly presented with the same input spike pattern? We show in this theoretical study that repeated inputs systematically lead to a shaping of the neuron's selectivity, emphasizing its very first input spikes, while steadily decreasing the postsynaptic response latency. This was obtained under various conditions of background noise, and even under conditions where spiking latencies and firing rates, or synchrony, provided conflicting informations. The key role of first spikes demonstrated here provides further support for models using a single wave of spikes to implement rapid neural processing.

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56. Rate and Synchrony in Feedforward Networks of Coincidence Detectors: Analytical Solution

Shawn Mikula, Ernst Niebur

https://doi.org/10.1162/0899766053429408 | pp. 881-902

We provide an analytical recurrent solution for the firing rates and cross-correlations of feedforward networks with arbitrary connectivity, excitatory or inhibitory, in response to steady-state spiking input to all neurons in the first network layer. Connections can go between any two layers as long as no loops are produced. Mean firing rates and pairwise cross-correlations of all input neurons can be chosen individually. We apply this method to study the propagation of rate and synchrony information through sample networks to address the current debate regarding the efficacy of rate codes versus temporal codes. Our results from applying the network solution to several examples support the following conclusions: (1) differential propagation efficacy of rate and synchrony to higher layers of a feedforward network is dependent on both network and input parameters, and (2) previous modeling and simulation studies exclusively supporting either rate or temporal coding must be reconsidered within the limited range of network and input parameters used. Our exact, analytical solution for feedforward networks of coincidence detectors should prove useful for further elucidating the efficacy and differential roles of rate and temporal codes in terms of different network and input parameter ranges.

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57. Independent Variable Time-Step Integration of Individual Neurons for Network Simulations

William W. Lytton, Michael L. Hines

https://doi.org/10.1162/0899766053429453 | pp. 903-921

Realistic neural networks involve the coexistence of stiff, coupled, continuous differential equations arising from the integrations of individual neurons, with the discrete events with delays used for modeling synaptic connections. We present here an integration method, the local variable time-step method (lvardt), that uses separate variable-step integrators for individual neurons in the network. Cells that are undergoing excitation tend to have small time steps, and cells that are at rest with little synaptic input tend to have large time steps. A synaptic input to a cell causes reinitialization of only that cell's integrator without affecting the integration of other cells. We illustrated the use of lvardt on three models: a worst-case synchronizing mutual-inhibition model, a best-case synfire chain model, and a more realistic thalamocortical network model.

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58. Synaptic Shot Noise and Conductance Fluctuations Affect the Membrane Voltage with Equal Significance

Magnus J. E. Richardson, Wulfram Gerstner

https://doi.org/10.1162/0899766053429444 | pp. 923-947

The subthreshold membrane voltage of a neuron in active cortical tissue is a fluctuating quantity with a distribution that reflects the firing statistics of the presynaptic population. It was recently found that conductance-based synaptic drive can lead to distributions with a significant skew. Here it is demonstrated that the underlying shot noise caused by Poissonian spike arrival also skews the membrane distribution, but

in the opposite sense. Using a perturbative method, we analyze the effects of shot noise on the distribution of synaptic conductances and calculate the consequent voltage distribution. To first order in the perturbation theory, the voltage distribution is a gaussian modulated by a prefactor that captures the skew. The gaussian component is identical to distributions derived using current-based models with an effective membrane time constant. The well-known effective-time-constant approximation can therefore be identified as the leading-order solution to the full conductance-based model. The higher-order modulatory prefactor containing the skew comprises terms due to both shot noise and conductance fluctuations. The diffusion approximation misses these shot-noise effects implying that analytical approaches such as the Fokker-Planck equation or simulation with filtered white noise cannot be used to improve on the gaussian approximation. It is further demonstrated that quantities used for fitting theory to experiment, such as the voltage mean and variance, are robust against these non-Gaussian effects. The effective-time-constant approximation is therefore relevant to experiment and provides a simple analytic base on which other pertinent biological details may be added.

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59. Dynamical Behaviors of a Large Class of General Delayed Neural Networks

Tianping Chen, Wenlian Lu, Guanrong Chen

https://doi.org/10.1162/0899766053429417 | pp. 949-968

Research of delayed neural networks with varying self-inhibitions, interconnection weights, and inputs is an important issue. In the real world, self-inhibitions, interconnection weights, and inputs should vary as time varies. In this letter, we discuss a large class of delayed neural networks with periodic inhibitions, interconnection weights, and inputs. We prove that if the activation functions are of Lipschitz type and some set of inequalities, for example, the set of inequalities 3.1 in theorem 1, is satisfied, the delayed system has a unique periodic solution, and any solution will converge to this periodic solution. We also prove that if either set of inequalities 3.20 in theorem 2 or 3.23 in theorem 3 is satisfied, then the system is exponentially stable globally. This class of delayed dynamical systems provides a general framework for many delayed dynamical systems. As special cases, it includes delayed Hopfield neural networks and cellular neural networks as well as distributed delayed neural networks with periodic self-inhibitions, interconnection weights, and inputs. Moreover, the entire discussion applies to delayed systems with constant self-inhibitions, interconnection weights, and inputs.

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60. A Subjective Distance Between Stimuli: Quantifying the Metric Structure of Representations

Damián Oliva, Inés Samengo, Stefan Leutgeb, Sheri J. Y. Mizumori

https://doi.org/10.1162/0899766053429372 | pp. 969-990

As subjects perceive the sensory world, different stimuli elicit a number of neural representations. Here, a subjective distance between stimuli is defined, measuring the degree of similarity between the underlying representations. As an example, the subjective distance between different locations in space is calculated from the activity of rodent's hippocampal place cells and lateral septal cells. Such a distance is compared to the real distance between locations. As the number of sampled neurons increases, the subjective distance shows a tendency to resemble the metrics of real space.

61. Gain Control by Concerted Changes in I_A and I_H Conductances

Denis Burdakov

https://doi.org/10.1162/0899766053491841 | pp. 991-995

Stability of intrinsic electrical activity and modulation of input-output gain are both important for neuronal information processing. It is therefore of interest to define biologically plausible parameters that allow these two features to coexist. Recent experiments indicate that in some biological neurons, the stability of spontaneous firing can arise from coregulated expression of the electrophysiologically opposing I(A) and I(H) currents. Here, I show that such balanced changes in I(A) and I(H) dramatically alter the slope of the relationship between the firing rate and driving current in a Hodgkin-Huxley-type model neuron. Concerted changes in I(A) and I(H) can thus control neuronal gain while preserving intrinsic activity.

62. Winner-Relaxing Self-Organizing Maps

Jens Christian Claussen

https://doi.org/10.1162/0899766053491922 | pp. 996-1009

A new family of self-organizing maps, the winner-relaxing Kohonen algorithm, is introduced as a generalization of a variant given by Kohonen in 1991. The magnification behavior is calculated analytically. For the original variant, a magnification exponent of 4/7 is derived; the generalized version allows steering the magnification in the wide range from exponent 1/2 to 1 in the one-dimensional case, thus providing optimal mapping in the sense of information theory. The winner-relaxing algorithm requires minimal extra computations per learning step and is conveniently easy to implement.

63. Image Segmentation by Networks of Spiking Neurons

Joachim M. Buhmann, Tilman Lange, Ulrich Ramacher

https://doi.org/10.1162/0899766053491913 | pp. 1010-1031

A network of leaky integrate-and-fire (IAF) neurons is proposed to segment gray-scale images. The network architecture with local competition between neurons that encode segment assignments of image blocks is motivated by a histogram clustering approach to image segmentation. Lateral excitatory connections between neighboring image sites yield a local smoothing of segments. The mean firing rate of class membership neurons encodes the image segmentation. A weight modification scheme is proposed that estimates segment-specific prototypical histograms. The robustness properties of the network implementation make it amenable to an analog VLSI realization. Results on synthetic and real-world images demonstrate the effectiveness of the architecture.

64. Neural Modeling of an Internal Clock

Tadashi Yamazaki, Shigeru Tanaka

https://doi.org/10.1162/0899766053491850 | pp. 1032-1058

We studied a simple random recurrent inhibitory network. Despite its simplicity, the dynamics was so rich that activity patterns of neurons evolved with time without recurrence due to random recurrent connections among neurons. The sequence of activity patterns was generated by the trigger of an external signal, and the generation was stable against noise. Moreover, the same sequence was reproducible using a strong transient signal, that is, the sequence generation could be reset. Therefore, a time passage from the trigger of an external signal could be represented by the sequence of activity patterns, suggesting that this model could work as an internal clock. The model could generate different sequences of activity patterns by providing different external signals; thus, spatiotemporal information could be represented by this model. Moreover, it was possible to speed up and slow down the sequence generation.

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65. Mirror Symmetric Topographic Maps Can Arise from Activity-Dependent Synaptic Changes

Reiner Schulz, James A. Reggia

https://doi.org/10.1162/0899766053491904 | pp. 1059-1083

Multiple adjacent, roughly mirror-image topographic maps are commonly observed in the sensory neocortex of many species. The cortical regions occupied by these maps are generally believed to be determined initially by genetically controlled chemical markers during development, with thalamocortical afferent activity subsequently exerting a progressively increasing influence over time. Here we use a computational model to show that adjacent topographic maps with mirror-image symmetry can arise from activity-dependent synaptic changes whenever the distribution radius of afferents sufficiently exceeds that of horizontal intracortical interactions. Which map edges become adjacent is strongly influenced by the probability distribution of input stimuli during map formation. Our results suggest that activity-dependent synaptic changes may play a role in influencing how adjacent maps become oriented following the initial establishment of cortical areas via genetically determined chemical markers. Further, the model unexpectedly predicts the occasional occurrence of adjacent maps with a different rotational symmetry. We speculate that such atypically oriented maps, in the context of otherwise normally interconnected cortical regions, might contribute to abnormal cortical information processing in some neurodevelopmental disorders.

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66. Stochastic Optimal Control and Estimation Methods Adapted to the Noise Characteristics of the Sensorimotor System

Emanuel Todorov

https://doi.org/10.1162/0899766053491887 | pp. 1084-1108

Optimality principles of biological movement are conceptually appealing and straightforward to formulate. Testing them empirically, however, requires the solution to stochastic optimal control and estimation problems for reasonably realistic models of the motor task and the sensorimotor periphery. Recent studies have highlighted the importance of incorporating biologically plausible noise into such models. Here we extend the linear-quadratic-gaussian framework--currently the only framework where such problems can be solved efficiently--to include control-dependent, state-dependent, and internal noise. Under this extended noise model, we derive a coordinate-descent algorithm guaranteed to converge to a feedback control law and a nonadaptive linear estimator optimal with respect to each other. Numerical simulations indicate that

convergence is exponential, local minima do not exist, and the restriction to nonadaptive linear estimators has negligible effects in the control problems of interest. The application of the algorithm is illustrated in the context of reaching movements. A Matlab implementation is available at www.cogsci.ucsd.edu/~todorov.

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67. Universal Approximation Capability of Cascade Correlation for Structures

Barbara Hammer, Alessio Micheli, Alessandro Sperduti

https://doi.org/10.1162/0899766053491878 | pp. 1109-1159

Cascade correlation (CC) constitutes a training method for neural networks that determines the weights as well as the neural architecture during training. Various extensions of CC to structured data have been proposed: recurrent cascade correlation (RCC) for sequences, recursive cascade correlation (RecCC) for tree structures with limited fan-out, and contextual recursive cascade correlation (CRecCC) for rooted directed positional acyclic graphs (DPAGs) with limited fan-in and fan-out. We show that these models possess the universal approximation property in the following sense: given a probability measure P on the input set, every measurable function from sequences into a real vector space can be approximated by a sigmoidal RCC up to any desired degree of accuracy up to inputs of arbitrary small probability. Every measurable function from tree structures with limited fan-out into a real vector space can be approximated by a sigmoidal RecCC with multiplicative neurons up to any desired degree of accuracy up to inputs of arbitrary small probability. For sigmoidal CRecCC networks with multiplicative neurons, we show the universal approximation capability for functions on an important subset of all DPAGs with limited fan-in and fan-out for which a specific linear representation yields unique codes. We give one sufficient structural condition for the latter property, which can easily be tested: the enumeration of ingoing and outgoing edges should becom patible. This property can be fulfilled for every DPAG with fan-in and fan-out two via reenumeration of children and parents, and for larger fan-in and fan-out via an expansion of the fan-in and fan-out and reenumeration of children and parents. In addition, the result can be generalized to the case of input-output isomorphic transductions of structures. Thus, CRecCC networks consti-tute the first neural models for which the universal approximation ca-pability of functions involving fairly general acyclic graph structures is proved.

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68. SVM Soft Margin Classifiers: Linear Programming versus Quadratic Programming

Qiang Wu, Ding■Xuan Zhou

https://doi.org/10.1162/0899766053491896 | pp. 1160-1187

Support vector machine (SVM) soft margin classifiers are important learning algorithms for classification problems. They can be stated as convex optimization problems and are suitable for a large data setting. Linear programming SVM classifiers are especially efficient for very large size samples. But little is known about their convergence, compared with the well-understood quadratic programming SVM classifier. In this article, we point out the difficulty and provide an error analysis. Our analysis shows that the convergence behavior of the linear programming SVM is almost the same as that of the quadratic programming SVM. This is implemented by setting a stepping-stone between the linear programming SVM and the classical 1-norm soft margin classifier. An upper bound for the misclassification error is presented for general probability distributions. Explicit learning rates are derived for deterministic and weakly separable

distributions, and for distributions satisfying some Tsybakov noise condition.

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69. Leave-One-Out Bounds for Support Vector Regression Model Selection

Ming■Wei Chang, Chih■Jen Lin

https://doi.org/10.1162/0899766053491869 | pp. 1188-1222

Minimizing bounds of leave-one-out errors is an important and efficient approach for support vector machine (SVM) model selection. Past research focuses on their use for classification but not regression. In this letter, we derive various leave-one-out bounds for support vector regression (SVR) and discuss the difference from those for classification. Experiments demonstrate that the proposed bounds are competitive with Bayesian SVR for parameter selection. We also discuss the differentiability of leave-one-out bounds.

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70. Rule Extraction from Recurrent Neural Networks: ATaxonomy and Review

Henrik Jacobsson

https://doi.org/10.1162/0899766053630350 | pp. 1223-1263

Rule extraction (RE) from recurrent neural networks (RNNs) refers to finding models of the underlying RNN, typically in the form of finite state machines, that mimic the network to a satisfactory degree while having the advantage of being more transparent. RE from RNNs can be argued to allow a deeper and more profound form of analysis of RNNs than other, more or less ad hoc methods. RE may give us understanding of RNNs in the intermediate levels between quite abstract theoretical knowledge of RNNs as a class of computing devices and quantitative performance evaluations of RNN instantiations. The development of techniques for extraction of rules from RNNs has been an active field since the early 1990s. This article reviews the progress of this development and analyzes it in detail. In order to structure the survey and evaluate the techniques, a taxonomy specifically designed for this purpose has been developed. Moreover, important open research issues are identified that, if addressed properly, possibly can give the field a significant push forward.

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71. Learning by Kernel Polarization

Yoram Baram

https://doi.org/10.1162/0899766053630341 | pp. 1264-1275

Kernels are key components of pattern recognition mechanisms. We propose a universal kernel optimality criterion, which is independent of the classifier to be used. Defining data polarization as a process by which points of different classes are driven to geometrically opposite locations in a confined domain, we propose selecting the kernel parameter values that polarize the data in the associated feature space. Conversely, the kernel is said to be polarized by the data. Kernel polarization gives rise to an unconstrained optimization problem. We show that complete kernel polarization yields consistent classification by kernel-sum classifiers. Tested on real-life data, polarized kernels demonstrate a clear advantage over the Euclidean distance in proximity classifiers. Embedded in a support vectors classifier, kernel polarization is found to yield about the same performance as exhaustive parameter search.

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72. A Unified Approach to Building and Controlling Spiking Attractor Networks

Chris Eliasmith

https://doi.org/10.1162/0899766053630332 | pp. 1276-1314

Extending work in Eliasmith and Anderson (2003), we employ a general framework to construct biologically plausible simulations of the three classes of attractor networks relevant for biological systems: static (point, line, ring, and plane) attractors, cyclic attractors, and chaotic attractors. We discuss these attractors in the context of the neural systems that they have been posited to help explain: eye control, working memory, and head direction; locomotion (specifically swimming); and olfaction, respectively. We then demonstrate how to introduce control into these models. The addition of control shows how attractor networks can be used as subsystems in larger neural systems, demonstrates how a much larger class of networks can be related to attractor networks, and makes it clear how attractor networks can be exploited for various information processing tasks in neurobiological systems.

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73. Synchronized Firings in the Networks of Class 1 Excitable Neurons with Excitatory and Inhibitory Connections and Their Dependences on the Forms of Interactions

Takashi Kanamaru, Masatoshi Sekine

https://doi.org/10.1162/0899766053630387 | pp. 1315-1338

Synchronized firings in the networks of class 1 excitable neurons with excitatory and inhibitory connections are investigated, and their dependences on the forms of interactions are analyzed. As the forms of interactions, we treat the double exponential coupling and the interactions derived from it: pulse coupling, exponential coupling, and alpha coupling. It is found that the bifurcation structure of the networks depends mainly on the decay time of the synaptic interaction and the effect of the rise time is smaller than that of the decay time.

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74. A Hierarchy of Associations in Hippocampo-Cortical Systems: Cognitive Maps and Navigation Strategies

J.P. Banquet, Ph. Gaussier, Mathias Quoy, A. Revel, Yves Burnod

https://doi.org/10.1162/0899766053630369 | pp. 1339-1384

In this letter we describe a hippocampo-cortical model of spatial processing and navigation based on a cascade of increasingly complex associative processes that are also relevant for other hippocampal functions such as episodic memory. Associative learning of different types and the related pattern encoding-recognition take place at three successive levels: (1) an object location level, which computes the landmarks from merged multimodal sensory inputs in the parahippocampal cortices; (2) a subject location level, which computes place fields by combination of local views and movement-related information in the entorhinal cortex; and (3) a spatiotemporal level, which computes place transitions from contiguous place fields in the CA3-CA1 region, which form building blocks for learning temporospatial sequences. At the cell population level, superficial entorhinal place cells encode spatial, context-independent maps as landscapes of activity; populations of transition cells in the CA3-CA1 region encode context-dependent maps as

sequences of transitions, which form graphs in prefrontal-parietal cortices. The model was tested on a robot moving in a real environment; these tests produced results that could help to interpret biological data. Two different goal-oriented navigation strategies were displayed depending on the type of map used by the system. Thanks to its multilevel, multimodal integration and behavioral implementation, the model suggests functional interpretations for largely unaccounted structural differences between hippocampo-cortical systems. Further, spatiotemporal information, a common denominator shared by several brain structures, could serve as a cognitive processing frame and a functional link, for example, during spatial navigation and episodic memory, as suggested by the applications of the model to other domains, temporal sequence learning and imitation in particular.

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75. Evidence Evaluation for Bayesian Neural Networks Using Contour Monte Carlo

Faming Liang

https://doi.org/10.1162/0899766053630323 | pp. 1385-1410

Bayesian neural networks play an increasingly important role in modeling and predicting nonlinear phenomena in scientific computing. In this article, we propose to use the contour Monte Carlo algorithm to evaluate evidence for Bayesian neural networks. In the new method, the evidence is dynamically learned for each of the models. Our numerical results show that the new method works well for both the regression and classification multilayer perceptrons. It often leads to an improved estimate, in terms of overall accuracy, for the evidence of multiple MLPs in comparison with the reversible-jump Markov chain Monte Carlo method and the gaussian approximation method. For the simulated data, it can identify the true models, and for the real data, it can produce results consistent with those published in the literature.

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76. An Ensemble of Cooperative Extended Kohonen Maps for Complex Robot Motion Tasks

Kian Hsiang Low, Wee Kheng Leow, Marcelo H. Ang

https://doi.org/10.1162/0899766053630378 | pp. 1411-1445

Self-organizing feature maps such as extended Kohonen maps (EKMs) have been very successful at learning sensorimotor control for mobile robot tasks. This letter presents a new ensemble approach, cooperative EKMs with indirect mapping, to achieve complex robot motion. An indirect-mapping EKM self-organizes to map from the sensory input space to the motor control space indirectly via a control parameter space. Quantitative evaluation reveals that indirect mapping can provide finer, smoother, and more efficient motion control than does direct mapping by operating in a continuous, rather than discrete, motor control space. It is also shown to outperform basis function neural networks. Furthermore, training its control parameters with recursive least squares enables faster convergence and better performance compared to gradient descent. The cooperation and competition of multiple self-organized EKMs allow a nonholonomic mobile robot to negotiate unforeseen, concave, closely spaced, and dynamic obstacles. Qualitative and quantitative comparisons with neural network ensembles employing weighted sum reveal that our method can achieve more sophisticated motion tasks even though the weighted-sum ensemble approach also operates in continuous motor control space.

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77. Categorization of Neural Excitability Using Threshold Models

Arnaud Tonnelier

https://doi.org/10.1162/0899766053723087 | pp. 1447-1455

A classification of spiking neurons according to the transition from quiescence to periodic firing of action potentials is commonly used. Nonbursting neurons are classified into two types, type I and type II excitability. We use simple phenomenological spiking neuron models to derive a criterion for the determination of the neural excitability based on the afterpotential following a spike. The crucial characteristic is the existence for type II model of a positive overshoot, that is, a delayed after depolarization, during the recovery process of the membrane potential. Our prediction is numerically tested using well-known type I and type II models including the Connor, Walter, & McKown (1977) model and the Hodgkin-Huxley (1952) model.

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78. Theory of the Snowflake Plot and Its Relations to Higher-Order Analysis Methods

Gabriela Czanner, Sonja Grün, Satish Iyengar

https://doi.org/10.1162/0899766053723041 | pp. 1456-1479

The snowflake plot is a scatter plot that displays relative timings of three neurons. It has had rather limited use since its introduction by Perkel, Gerstein, Smith, and Tatton (1975), in part because its triangular coordinates are unfamiliar and its theoretical properties are not well studied. In this letter, we study certain quantitative properties of this plot: we use projections to relate the snowflake plot to the cross-correlation histogram and the spike-triggered joint histogram, study the sampling properties of the plot for the null case of independent spike trains, study a simulation of a coincidence detector, and describe the extension of this plot to more than three neurons.

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79. Asymptotic Theory of Information-Theoretic Experimental Design

Liam Paninski

https://doi.org/10.1162/0899766053723032 | pp. 1480-1507

We discuss an idea for collecting data in a relatively efficient manner. Our point of view is Bayesian and information-theoretic: on any given trial, we want to adaptively choose the input in such a way that the mutual information between the (unknown) state of the system and the (stochastic) output is maximal, given any prior information (including data collected on any previous trials). We prove a theorem that quantifies the effectiveness of this strategy and give a few illustrative examples comparing the performance of this adaptive technique to that of the more usual nonadaptive experimental design. In particular, we calculate the asymptotic efficiency of the information-maximization strategy and demonstrate that this method is in a well-defined sense never less efficient--and is generically more efficient--than the nonadaptive strategy. For example, we are able to explicitly calculate the asymptotic relative efficiency of the staircase method widely employed in psychophysics research and to demonstrate the dependence of this efficiency on the form of the psychometric function underlying the output responses.

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80. Maximum Likelihood Set for Estimating a Probability Mass Function

Bruno Jedynak, Sanjeev Khudanpur

https://doi.org/10.1162/0899766053723078 | pp. 1508-1530

We propose a new method for estimating the probability mass function (pmf) of a discrete and finite random variable from a small sample. We focus on the observed counts--the number of times each value appears in the sample--and define the maximum likelihood set (MLS) as the set of pmfs that put more mass on the observed counts than on any other set of counts possible for the same sample size. We characterize the MLS in detail in this article. We show that the MLS is a diamond-shaped subset of the probability simplex [0,1]k bounded by at most k x (k-1) hyper-planes, where k is the number of possible values of the random variable. The MLS always contains the empirical distribution, as well as a family of Bayesian estimators based on a Dirichlet prior, particularly the well-known Laplace estimator. We propose to select from the MLS the pmf that is closest to a fixed pmf that encodes prior knowledge. When using Kullback-Leibler distance for this selection, the optimization problem comprises finding the minimum of a convex function over a domain defined by linear inequalities, for which standard numerical procedures are available. We apply this estimate to language modeling using Zipf's law to encode prior knowledge and show that this method permits obtaining state-of-the-art results while being conceptually simpler than most competing methods.

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81. Estimating Entropy Rates with Bayesian Confidence Intervals

Matthew B. Kennel, Jonathon Shlens, Henry D. I. Abarbanel, E. J. Chichilnisky

https://doi.org/10.1162/0899766053723050 | pp. 1531-1576

The entropy rate quantifies the amount of uncertainty or disorder produced by any dynamical system. In a spiking neuron, this uncertainty translates into the amount of information potentially encoded and thus the subject of intense theoretical and experimental investigation. Estimating this quantity in observed, experimental data is difficult and requires a judicious selection of probabilistic models, balancing between two opposing biases. We use a model weighting principle originally developed for lossless data compression, following the minimum description length principle. This weighting yields a direct estimator of the entropy rate, which, compared to existing methods, exhibits significantly less bias and converges faster in simulation. With Monte Carlo techinques, we estimate a Bayesian confidence interval for the entropy rate. In related work, we apply these ideas to estimate the information rates between sensory stimuli and neural responses in experimental data (Shlens, Kennel, Abarbanel, & Chichilnisky, in preparation).

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82. Spike Timing Precision and Neural Error Correction: Local Behavior

Michael Stiber

https://doi.org/10.1162/0899766053723069 | pp. 1577-1601

The effects of spike timing precision and dynamical behavior on error correction in spiking neurons were investigated. Stationary discharges-phase locked, quasiperiodic, or chaotic-were induced in a simulated neuron by presenting pacemaker presynaptic spike trains across a model of a prototypical inhibitory synapse. Reduced timing precision was modeled by jittering presynaptic spike times. Aftereffects of errors-in this communication, missed presynaptic spikes-were determined by comparing postsynaptic spike times between simulations identical except for the presence or absence of errors. Results show that the effects of

an error vary greatly depending on the ongoing dynamical behavior. In the case of phase lockings, a high degree of presynaptic spike timing precision can provide significantly faster error recovery. For nonlocked behaviors, isolated missed spikes can have little or no discernible aftereffects (or even serve to paradoxically reduce uncertainty in postsynaptic spike timing), regardless of presynaptic imprecision. This suggests two possible categories of error correction: high-precision locking with rapid recovery and low-precision nonlocked with error immunity.

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83. A New Approach to Spatial Covariance Modeling of Functional Brain Imaging Data: Ordinal Trend Analysis

Christian Habeck, John W. Krakauer, Claude Ghez, Harold A. Sackeïm, David Eidelberg, Yaakov Stern, James R. Moeller

https://doi.org/10.1162/0899766053723023 | pp. 1602-1645

In neuroimaging studies of human cognitive abilities, brain activation patterns that include regions that are strongly interactive in response to experimental task demands are of particular interest. Among the existing network analyses, partial least squares (PLS; McIntosh, 1999; McIntosh, Bookstein, Haxby, & Grady, 1996) has been highly successful, particularly in identifying group differences in regional functional connectivity, including differences as diverse as those associated with states of awareness and normal aging. However, we address the need for a within-group model that identifies patterns of regional functional connectivity that exhibit sustained activity across graduated changes in task parameters. For example, predictions of sustained connectivity are commonplace in studies of cognition that involve a series of tasks over which task difficulty increases (Baddeley, 2003). We designed ordinal trend analysis (OrT) to identify activation patterns that increase monotonically in their expression as the experimental task parameter increases, while the correlative relationships between brain regions remain constant. Of specific interest are patterns that express positive ordinal trends on a subject-by-subject basis. A unique feature of OrT is that it recovers information about functional connectivity based solely on experimental design variables. In particular, there is no requirement by OrT to provide either a quantitative model of the uncertain relationship between functional brain circuitry and subject variables (e.g., task performance and IQ) or partial information about the regions that are functionally connected. In this letter, we provide a step-by-step recipe of the computations performed in the new OrT analysis, including a description of the inferential statistical methods applied. Second, we describe applications of OrT to an event-related fMRI study of verbal working memory and H(2)15O-PET study of visuo-motor learning. In sum, OrT has potential applications to not only studies of young adults and their cognitive abilities, but also studies of normal aging and neurological and psychiatric disease.

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84. Investigating the Fault Tolerance of Neural Networks

Elko B. Tchernev, Rory G. Mulvaney, Dhananjay S. Phatak

https://doi.org/10.1162/0899766053723096 | pp. 1646-1664

Particular levels of partial fault tolerance (PFT) in feedforward artificial neural networks of a given size can be obtained by redundancy (replicating a smaller normally trained network), by design (training specifically to increase PFT), and by a combination of the two (replicating a smaller PFT-trained network). This letter investigates the method of achieving the highest PFT per network size (total number of units and

connections) for classification problems. It concludes that for non-toy problems, there exists a normally trained network of optimal size that produces the smallest fully fault-tolerant network when replicated. In addition, it shows that for particular network sizes, the best level of PFT is achieved by training a network of that size for fault tolerance. The results and discussion demonstrate how the outcome depends on the levels of saturation of the network nodes when classifying data points. With simple training tasks, where the complexity of the problem and the size of the network are well within the ability of the training method, the hidden-layer nodes operate close to their saturation points, and classification is clean. Under such circumstances, replicating the smallest normally trained correct network yields the highest PFT for any given network size. For hard training tasks (difficult classification problems or network sizes close to the minimum), normal training obtains networks that do not operate close to their saturation points, and outputs are not as close to their targets. In this case, training a larger network for fault tolerance yields better PFT than replicating a smaller, normally trained network. However, since fault-tolerant training on its own produces networks that operate closer to their linear areas than normal training, replicating normally trained networks ultimately leads to better PFT than replicating fault-tolerant networks of the same initial size.

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85. How Close Are We to Understanding V1?

Bruno A. Olshausen, David J. Field

https://doi.org/10.1162/0899766054026639 | pp. 1665-1699

A wide variety of papers have reviewed what is known about the function of primary visual cortex. In this review, rather than stating what is known, we attempt to estimate how much is still unknown about V1 function. In particular, we identify five problems with the current view of V1 that stem largely from experimental and theoretical biases, in addition to the contributions of nonlinearities in the cortex that are not well understood. Our purpose is to open the door to new theories, a number of which we describe, along with some proposals for testing them.

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86. Motion Contrast Classification Is a Linearly Nonseparable Problem

Alireza S. Mahani, Ralf Weßel

https://doi.org/10.1162/0899766054026666 | pp. 1700-1705

Sensitivity to image motion contrast, that is, the relative motion between different parts of the visual field, is a common and computationally important property of many neurons in the visual pathways of vertebrates. Here we illustrate that, as a classification problem, motion contrast detection is linearly nonseparable. In order to do so, we prove a theorem stating a sufficient condition for linear nonseparability. We argue that nonlinear combinations of local measurements of velocity at different locations and times are needed in order to solve the motion contrast problem.

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87. Edgeworth-Expanded Gaussian Mixture Density Modeling

Marc M. Van Hulle

https://doi.org/10.1162/0899766054026657 | pp. 1706-1714

Instead of increasing the order of the Edgeworth expansion of a single gaussian kernel, we suggest using mixtures of Edgeworth-expanded gaussian kernels of moderate order. We introduce a simple closed-form solution for estimating the kernel parameters based on weighted moment matching. Furthermore, we formulate the extension to the multivariate case, which is not always feasible with algebraic density approximation procedures.

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88. Movement Generation with Circuits of Spiking Neurons

Prashant Joshi, Wolfgang Maass

https://doi.org/10.1162/0899766054026684 | pp. 1715-1738

How can complex movements that take hundreds of milliseconds be generated by stereotypical neural microcircuits consisting of spiking neurons with a much faster dynamics? We show that linear readouts from generic neural microcircuit models can be trained to generate basic arm movements. Such movement generation is independent of the arm model used and the type of feedback that the circuit receives. We demonstrate this by considering two different models of a two-jointed arm, a standard model from robotics and a standard model from biology, that each generates different kinds of feedback. Feedback that arrives with biologically realistic delays of 50 to 280 ms turns out to give rise to the best performance. If a feedback with such desirable delay is not available, the neural microcircuit model also achieves good performance if it uses internally generated estimates of such feedback. Existing methods for movement generation in robotics that take the particular dynamics of sensors and actuators into account (embodiment of motor systems) are taken one step further with this approach, which provides methods for also using the embodiment of motion generation circuitry, that is, the inherent dynamics and spatial structure of neural circuits, for the generation of movement.

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89. Cognitive Enhancement Mediated Through Postsynaptic Actions of Norepinephrine on Ongoing Cortical Activity

Osamu Hoshino

https://doi.org/10.1162/0899766054026675 | pp. 1739-1775

We propose two distinct types of norepinephrine (NE)-neuromodulatory systems: an enhanced-excitatory and enhanced-inhibitory (E-E/E-I) system and a depressed-excitatory and enhanced-inhibitory (D-E/E-I) system. In both systems, inhibitory synaptic efficacies are enhanced, but excitatory ones are modified in a contradictory manner: the E-E/E-I system enhances excitatory synaptic efficacies, whereas the D-E/E-I system depresses them. The E-E/E-I and D-E/E-I systems altered the dynamic property of ongoing (background) neuronal activity and greatly influenced the cognitive performance (S/N ratio) of a cortical neural network. The E-E/E-I system effectively enhanced S/N ratio for weaker stimuli with lower doses of NE, whereas the D-E/E-I system enhanced stronger stimuli with higher doses of NE. The neural network effectively responded to weaker stimuli if brief gamma-bursts were involved in ongoing neuronal activity that is controlled under the E-E/E-I neuromodulatory system. If the E-E/E-I and the D-E/E-I systems interact within the neural network, depressed neurons whose activity is depressed by NE application have bimodal property. That is, S/N ratio can be enhanced not only for stronger stimuli as its original property but also for weaker stimuli, for which coincidental neuronal firings among enhanced neurons whose activity is enhanced by NE application are essential. We suggest that the recruitment of the depressed neurons for the detection

of weaker (subthreshold) stimuli might be advantageous for the brain to cope with a variety of sensory stimuli.

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90. Advancing the Boundaries of High-Connectivity Network Simulation with Distributed Computing

Abigail Morrison, Carsten Mehring, T. Geisel, Ad Aertsen, Markus Diesmann

https://doi.org/10.1162/0899766054026648 | pp. 1776-1801

The availability of efficient and reliable simulation tools is one of the mission-critical technologies in the fast-moving field of computational neuroscience. Research indicates that higher brain functions emerge from large and complex cortical networks and their interactions. The large number of elements (neurons) combined with the high connectivity (synapses) of the biological network and the specific type of interactions impose severe constraints on the explorable system size that previously have been hard to overcome. Here we present a collection of new techniques combined to a coherent simulation tool removing the fundamental obstacle in the computational study of biological neural networks: the enormous number of synaptic contacts per neuron. Distributing an individual simulation over multiple computers enables the investigation of networks orders of magnitude larger than previously possible. The software scales excellently on a wide range of tested hardware, so it can be used in an interactive and iterative fashion for the development of ideas, and results can be produced quickly even for very large networks. In contrast to earlier approaches, a wide class of neuron models and synaptic dynamics can be represented.

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91. Dynamical Analysis of Continuous Higher-Order Hopfield Networks for Combinatorial Optimization

Miguel Atencia, Gonzalo Joya, F. Sandoval

https://doi.org/10.1162/0899766054026620 | pp. 1802-1819

In this letter, the ability of higher-order Hopfield networks to solve combinatorial optimization problems is assessed by means of a rigorous analysis of their properties. The stability of the continuous network is almost completely clarified: (1) hyperbolic interior equilibria, which are unfeasible, are unstable; (2) the state cannot escape from the unitary hypercube; and (3) a Lyapunov function exists. Numerical methods used to implement the continuous equation on a computer should be designed with the aim of preserving these favorable properties. The case of nonhyperbolic fixed points, which occur when the Hessian of the target function is the null matrix, requires further study. We prove that these nonhyperbolic interior fixed points are unstable in networks with three neurons and order two. The conjecture that interior equilibria are unstable in the general case is left open.

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92. Some Generalized Sufficient Convergence Criteria for Nonlinear Continuous Neural Networks

Jito Vanualailai, Shin-ichi Nakagiri

https://doi.org/10.1162/0899766054026701 | pp. 1820-1835

A reason for applying the direct method of Lyapunov to artificial neural networks (ANNs) is to design dynamical neural networks so that they exhibit global asymptotic stability. Lyapunov functions that frequently appear in the ANN literature include the quadratic function, the Persidskii function, and the Luré-Postnikov function. This contribution revisits the quadratic function and shows that via Krasovskii-like stability criteria, it is possible to have a very simple and systematic procedure to obtain not only new and generalized results but also well-known sufficient conditions for convergence established recently by non-Lyapunov methods, such as the matrix measure and nonlinear measure.

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93. Estimation and Marginalization Using the Kikuchi Approximation Methods

Payam Pakzad, Venkat Anantharam

https://doi.org/10.1162/0899766054026693 | pp. 1836-1873

In this letter, we examine a general method of approximation, known as the Kikuchi approximation method, for finding the marginals of a product distribution, as well as the corresponding partition function. The Kikuchi approximation method defines a certain constrained optimization problem, called the Kikuchi problem, and treats its stationary points as approximations to the desired marginals. We show how to associate a graph to any Kikuchi problem and describe a class of local message-passing algorithms along the edges of any such graph, which attempt to find the solutions to the problem. Implementation of these algorithms on graphs with fewer edges requires fewer operations in each iteration. We therefore characterize minimal graphs for a Kikuchi problem, which are those with the minimum number of edges. We show with empirical results that these simpler algorithms often offer significant savings in computational complexity, without suffering a loss in the convergence rate. We give conditions for the convexity of a given Kikuchi problem and the exactness of the approximations in terms of the loops of the minimal graph. More precisely, we show that if the minimal graph is cycle free, then the Kikuchi approximation method is exact, and the converse is also true generically. Together with the fact that in the cycle-free case, the iterative algorithms are equivalent to the well-known belief propagation algorithm, our results imply that, generically, the Kikuchi approximation method can be exact if and only if traditional junction tree methods could also solve the problem exactly.

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94. The Cocktail Party Problem

S. Haykin, Zhe Chen

https://doi.org/10.1162/0899766054322964 | pp. 1875-1902

This review presents an overview of a challenging problem in auditory perception, the cocktail party phenomenon, the delineation of which goes back to a classic paper by Cherry in 1953. In this review, we address the following issues: (1) human auditory scene analysis, which is a general process carried out by the auditory system of a human listener; (2) insight into auditory perception, which is derived from Marr's vision theory; (3) computational auditory scene analysis, which focuses on specific approaches aimed at solving the machine cocktail party problem; (4) active audition, the proposal for which is motivated by analogy with active vision, and (5) discussion of brain theory and independent component analysis, on the one hand, and correlative neural firing, on the other.

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95. Edgeworth Approximation of Multivariate Differential Entropy

Marc M. Van Hulle
https://doi.org/10.1162/0899766054323026 | pp. 1903-1910

We develop the general, multivariate case of the Edgeworth approximation of differential entropy and show that it can be more accurate than the nearest-neighbor method in the multivariate case and that it scales better with sample size. Furthermore, we introduce mutual information estimation as an application.

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96. Perfect Fault Tolerance of the n-k-n Network

Elko B. Tchernev, Rory G. Mulvaney, Dhananjay S. Phatak
https://doi.org/10.1162/0899766054322946 | pp. 1911-1920

It was shown in Phatak, Choi, and Koren (1993) that the neural network implementation of the n-k-n encoder-decoder has a minimal size of k = 2, and it was shown how to construct the network. A proof was given in Phatak and Koren (1995) that in order to achieve perfect fault tolerance by replicating the hidden layer, the required number of replications is at least three. In this note, exact lower bounds are derived for the number of replications as a function of n, for the n-2-n network and for the n- log2n-n network.

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97. On the Slow Convergence of EM and VBEM in Low-Noise Linear Models

Kaare Brandt Petersen, Ole Winther, Lars Kai Hansen
https://doi.org/10.1162/0899766054322991 | pp. 1921-1926

We analyze convergence of the expectation maximization (EM) and variational Bayes EM (VBEM) schemes for parameter estimation in noisy linear models. The analysis shows that both schemes are inefficient in the low-noise limit. The linear model with additive noise includes as special cases independent component analysis, probabilistic principal component analysis, factor analysis, and Kalman filtering. Hence, the results are relevant for many practical applications.

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98. Analyzing Functional Connectivity Using a Network Likelihood Model of Ensemble Neural Spiking Activity

Murat Okatan, Matthew A. Wilson, Emery N. Brown
https://doi.org/10.1162/0899766054322973 | pp. 1927-1961

Analyzing the dependencies between spike trains is an important step in understanding how neurons work in concert to represent biological signals. Usually this is done for pairs of neurons at a time using correlation-based techniques. Chornoboy, Schramm, and Karr (1988) proposed maximum likelihood methods for the simultaneous analysis of multiple pair-wise interactions among an ensemble of neurons. One of these methods is an iterative, continuous-time estimation algorithm for a network likelihood model formulated in terms of multiplicative conditional intensity functions. We devised a discrete-time version of this algorithm that includes a new, efficient computational strategy, a principled method to compute starting values, and a principled stopping criterion. In an analysis of simulated neural spike trains from ensembles of interacting neurons, the algorithm recovered the correct connectivity matrices and interaction parameters. In the analysis of spike trains from an ensemble of rat hippocampal place cells, the algorithm identified a

connectivity matrix and interaction parameters consistent with the pattern of conjoined firing predicted by the overlap of the neurons' spatial receptive fields. These results suggest that the network likelihood model can be an efficient tool for the analysis of ensemble spiking activity.

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99. Data-Robust Tight Lower Bounds to the Information Carried by Spike Times of a Neuronal Population

Giordano Pola, Rasmus S. Petersen, Alexander Thiele, M. P. Young, Stefano Panzeri

https://doi.org/10.1162/0899766054322955 | pp. 1962-2005

We develop new data-robust lower-bound methods to quantify the information carried by the timing of spikes emitted by neuronal populations. These methods have better sampling properties and are tighter than previous bounds based on neglecting correlation in the noise entropy. Our new lower bounds are precise also in the presence of strongly correlated firing. They are not precise only if correlations are strongly stimulus modulated over a long time range. Under conditions typical of many neurophysiological experiments, these techniques permit precise information estimates to be made even with data samples that are three orders of magnitude smaller than the size of the response space.

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100. Fluctuation-Dissipation Theorem and Models of Learning

Ilya Nemenman

https://doi.org/10.1162/0899766054322982 | pp. 2006-2033

Advances in statistical learning theory have resulted in a multitude of different designs of learning machines. But which ones are implemented by brains and other biological information processors? We analyze how various abstract Bayesian learners perform on different data and argue that it is difficult to determine which learning-theoretic computation is performed by a particular organism using just its performance in learning a stationary target (learning curve). Based on the fluctuation-dissipation relation in statistical physics, we then discuss a different experimental setup that might be able to solve the problem.

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101. Correlated Firing in a Feedforward Network with Mexican-Hat-Type Connectivity

Kosuke Hamaguchi, Masato Okada, Michiko Yamana, Kazuyuki Aihara

https://doi.org/10.1162/0899766054322937 | pp. 2034-2059

We report on deterministic and stochastic evolutions of firing states through a feedforward neural network with Mexican-hat-type connectivity. The prevalence of columnar structures in a cortex implies spatially localized connectivity between neural pools. Although feedforward neural network models with homogeneous connectivity have been intensively studied within the context of the synfire chain, the effect of local connectivity has not yet been studied so thoroughly. When a neuron fires independently, the dynamics of macroscopic state variables (a firing rate and spatial eccentricity of a firing pattern) is deterministic from the law of large numbers. Possible stable firing states, which are derived from deterministic evolution equations, are uniform, localized, and nonfiring. The multistability of these three states is obtained where the excitatory and inhibitory interactions among neurons are balanced. When the presynapse-dependent

variance in connection efficacies is incorporated into the network, the variance generates common noise. Then the evolution of the macroscopic state variables becomes stochastic, and neurons begin to fire in a correlated manner due to the common noise. The correlation structure that is generated by common noise exhibits a nontrivial bimodal distribution. The development of a firing state through neural layers does not converge to a certain fixed point but keeps on fluctuating.

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102. Supervised Learning in a Recurrent Network of Rate-Model Neurons Exhibiting Frequency Adaptation

Pierre A. Fortier, Emmanuel Guigon, Yves Burnod

https://doi.org/10.1162/0899766054323017 | pp. 2060-2076

For gradient descent learning to yield connectivity consistent with real biological networks, the simulated neurons would have to include more realistic intrinsic properties such as frequency adaptation. However, gradient descent learning cannot be used straightforwardly with adapting rate-model neurons because the derivative of the activation function depends on the activation history. The objectives of this study were to (1) develop a simple computational approach to reproduce mathematical gradient descent and (2) use this computational approach to provide supervised learning in a network formed of rate-model neurons that exhibit frequency adaptation. The results of mathematical gradient descent were used as a reference in evaluating the performance of the computational approach. For this comparison, standard (nonadapting) rate-model neurons were used for both approaches. The only difference was the gradient calculation: the mathematical approach used the derivative at a point in weight space, while the computational approach used the slope for a step change in weight space. Theoretically, the results of the computational approach should match those of the mathematical approach, as the step size is reduced but floating-point accuracy formed a lower limit to usable step sizes. A systematic search for an optimal step size yielded a computational approach that faithfully reproduced the results of mathematical gradient descent. The computational approach was then used for supervised learning of both connection weights and intrinsic properties of rate-model neurons to convert a tonic input into a phasic-tonic output pattern. Learning produced biologically realistic connectivity that essentially used a monosynaptic connection from the tonic input neuron to an output neuron with strong frequency adaptation as compared to a complex network when using nonadapting neurons. Thus, more biologically realistic connectivity was achieved by implementing rate-model neurons with more realistic intrinsic properties. Our computational approach could be applied to learning of other neuron properties.

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103. Learning Bounds for Kernel Regression Using Effective Data Dimensionality

Tong Zhang

https://doi.org/10.1162/0899766054323008 | pp. 2077-2098

Kernel methods can embed finite-dimensional data into infinite-dimensional feature spaces. In spite of the large underlying feature dimensionality, kernel methods can achieve good generalization ability. This observation is often wrongly interpreted, and it has been used to argue that kernel learning can magically avoid the "curse-of-dimensionality" phenomenon encountered in statistical estimation problems. This letter shows that although using kernel representation, one can embed data into an infinite-dimensional feature

space; the effective dimensionality of this embedding, which determines the learning complexity of the underlying kernel machine, is usually small. In particular, we introduce an algebraic definition of a scale-sensitive effective dimension associated with a kernel representation. Based on this quantity, we derive upper bounds on the generalization performance of some kernel regression methods. Moreover, we show that the resulting convergent rates are optimal under various circumstances.

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104. Gradient-Based Adaptation of General Gaussian Kernels

Tobias Glasmachers, Christian Igel

https://doi.org/10.1162/0899766054615635 | pp. 2099-2105

Gradient-based optimizing of gaussian kernel functions is considered. The gradient for the adaptation of scaling and rotation of the input space is computed to achieve invariance against linear transformations. This is done by using the exponential map as a parameterization of the kernel parameter manifold. By restricting the optimization to a constant trace subspace, the kernel size can be controlled. This is, for example, useful to prevent overfitting when minimizing radius-margin generalization performance measures. The concepts are demonstrated by training hard margin support vector machines on toy data.

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105. Learning Only When Necessary: Better Memories of Correlated Patterns in Networks with Bounded Synapses

Walter Senn, Stefano Fusi

https://doi.org/10.1162/0899766054615644 | pp. 2106-2138

Learning in a neuronal network is often thought of as a linear superposition of synaptic modifications induced by individual stimuli. However, since biological synapses are naturally bounded, a linear superposition would cause fast forgetting of previously acquired memories. Here we show that this forgetting can be avoided by introducing additional constraints on the synaptic and neural dynamics. We consider Hebbian plasticity of excitatory synapses. A synapse is modified only if the postsynaptic response does not match the desired output. With this learning rule, the original memory performances with unbounded weights are regained, provided that (1) there is some global inhibition, (2) the learning rate is small, and (3) the neurons can discriminate small differences in the total synaptic input (e.g., by making the neuronal threshold small compared to the total postsynaptic input). We prove in the form of a generalized perceptron convergence theorem that under these constraints, a neuron learns to classify any linearly separable set of patterns, including a wide class of highly correlated random patterns. During the learning process, excitation becomes roughly balanced by inhibition, and the neuron classifies the patterns on the basis of small differences around this balance. The fact that synapses saturate has the additional benefit that nonlinearly separable patterns, such as similar patterns with contradicting outputs, eventually generate a subthreshold response, and therefore silence neurons that cannot provide any information.

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106. Coding of Temporally Varying Signals in Networks of Spiking Neurons with Global Delayed Feedback

Naoki Masuda, Brent Doiron, André Longtin, Kazuyuki Aihara

https://doi.org/10.1162/0899766054615680 | pp. 2139-2175

Oscillatory and synchronized neural activities are commonly found in the brain, and evidence suggests that many of them are caused by global feedback. Their mechanisms and roles in information processing have been discussed often using purely feedforward networks or recurrent networks with constant inputs. On the other hand, real recurrent neural networks are abundant and continually receive information-rich inputs from the outside environment or other parts of the brain. We examine how feedforward networks of spiking neurons with delayed global feedback process information about temporally changing inputs. We show that the network behavior is more synchronous as well as more correlated with and phase-locked to the stimulus when the stimulus frequency is resonant with the inherent frequency of the neuron or that of the network oscillation generated by the feedback architecture. The two eigenmodes have distinct dynamical characteristics, which are supported by numerical simulations and by analytical arguments based on frequency response and bifurcation theory. This distinction is similar to the class I versus class II classification of single neurons according to the bifurcation from quiescence to periodic firing, and the two modes depend differently on system parameters. These two mechanisms may be associated with different types of information processing.

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107. Attention-Gated Reinforcement Learning of Internal Representations for Classification

Pieter R. Roelfsema, Arjen van Ooyen

https://doi.org/10.1162/0899766054615699 | pp. 2176-2214

Animal learning is associated with changes in the efficacy of connections between neurons. The rules that govern this plasticity can be tested in neural networks. Rules that train neural networks to map stimuli onto outputs are given by supervised learning and reinforcement learning theories. Supervised learning is efficient but biologically implausible. In contrast, reinforcement learning is biologically plausible but comparatively inefficient. It lacks a mechanism that can identify units at early processing levels that play a decisive role in the stimulus-response mapping. Here we show that this so-called credit assignment problem can be solved by a new role for attention in learning. There are two factors in our new learning scheme that determine synaptic plasticity: (1) a reinforcement signal that is homogeneous across the network and depends on the amount of reward obtained after a trial, and (2) an attentional feedback signal from the output layer that limits plasticity to those units at earlier processing levels that are crucial for the stimulus-response mapping. The new scheme is called attention-gated reinforcement learning (AGREL). We show that it is as efficient as supervised learning in classification tasks. AGREL is biologically realistic and integrates the role of feedback connections, attention effects, synaptic plasticity, and reinforcement learning signals into a coherent framework.

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108. Computing with Continuous Attractors: Stability and Online Aspects

Si Wu, ■■■-■■■ ■■■■■

https://doi.org/10.1162/0899766054615626 | pp. 2215-2239

Two issues concerning the application of continuous attractors in neural systems are investigated: the computational robustness of continuous attractors with respect to input noises and the implementation of

Bayesian online decoding. In a perfect mathematical model for continuous attractors, decoding results for stimuli are highly sensitive to input noises, and this sensitivity is the inevitable consequence of the system's neutral stability. To overcome this shortcoming, we modify the conventional network model by including extra dynamical interactions between neurons. These interactions vary according to the biologically plausible Hebbian learning rule and have the computational role of memorizing and propagating stimulus information accumulated with time. As a result, the new network model responds to the history of external inputs over a period of time, and hence becomes insensitive to short-term fluctuations. Also, since dynamical interactions provide a mechanism to convey the prior knowledge of stimulus, that is, the information of the stimulus presented previously, the network effectively implements online Bayesian inference. This study also reveals some interesting behavior in neural population coding, such as the trade-off between decoding stability and the speed of tracking time-varying stimuli, and the relationship between neural tuning width and the tracking speed.

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109. Optimal Signal Estimation in Neuronal Models

Petr Lánský, Priscilla E. Greenwood

https://doi.org/10.1162/0899766054615653 | pp. 2240-2257

We study optimal estimation of a signal in parametric neuronal models on the basis of interspike interval data. Fisher information is the inverse asymptotic variance of the best estimator. Its dependence on the parameter value indicates accuracy of estimation. Our models assume that the input signal is estimated from neuronal output interspike interval data where the frequency transfer function is sigmoidal. If the coefficient of variation of the interspike interval is constant with respect to the signal, the Fisher information is unimodal, and its maximum for the most estimable signal can be found. We obtain a general result and compare the signal producing maximal Fisher information with the inflection point of the sigmoidal transfer function in several basic neuronal models.

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110. Finite State Automata Resulting from Temporal Information Maximization and a Temporal Learning Rule

Thomas Wennekers, Nihat Ay

https://doi.org/10.1162/0899766054615671 | pp. 2258-2290

We extend Linkser's Infomax principle for feedforward neural networks to a measure for stochastic interdependence that captures spatial and temporal signal properties in recurrent systems. This measure, stochastic interaction, quantifies the Kullback-Leibler divergence of a Markov chain from a product of split chains for the single unit processes. For unconstrained Markov chains, the maximization of stochastic interaction, also called Temporal Infomax, has been previously shown to result in almost deterministic dynamics. This letter considers Temporal Infomax on constrained Markov chains, where some of the units are clamped to prescribed stochastic processes providing input to the system. Temporal Infomax in that case leads to finite state automata, either completely deterministic or weakly nondeterministic. Transitions between internal states of these systems are almost perfectly predictable given the complete current state and the input, but the activity of each single unit alone is virtually random. The results are demonstrated by means of computer simulations and confirmed analytically. It is furthermore shown numerically that Temporal Infomax leads to a high information flow from the input to internal units and that a simple temporal

learning rule can approximately achieve the optimization of temporal interaction. We relate these results to experimental data concerning the correlation dynamics and functional connectivities observed in multiple electrode recordings.

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111. Fast Access to Concepts in Concept Lattices via Bidirectional Associative Memories

Rohana K. Rajapakse, M.J. Denham

https://doi.org/10.1162/0899766054615662 | pp. 2291-2300

Bidirectional associative memories (BAMs) are shown to be capable of precisely learning concept lattice structures by Radim Belohlávek. The focus of this letter is to show that the BAM, when set up with a concept lattice by setting up connection weights according to the rule proposed by Belohlávek, always returns the most specific or most generic concept containing the given set of objects or attributes when a set of objects or attributes is presented as input to the object or attribute layer. A proof of this property is given here, together with an example, and a brief application of the property is provided.

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112. An Extended Analytic Expression for the Membrane Potential Distribution of Conductance-Based Synaptic Noise

Michelle Rudolph, Alain Destexhe

https://doi.org/10.1162/0899766054796932 | pp. 2301-2315

Synaptically generated subthreshold membrane potential (Vm) fluctuations can be characterized within the framework of stochastic calculus. It is possible to obtain analytic expressions for the steady-state Vm distribution, even in the case of conductance-based synaptic currents. However, as we show here, the analytic expressions obtained may substantially deviate from numerical solutions if the stochastic membrane equations are solved exclusively based on expectation values of differentials of the stochastic variables, hence neglecting the spectral properties of the underlying stochastic processes. We suggest a simple solution that corrects these deviations, leading to extended analytic expressions of the Vm distribution valid for a parameter regime that covers several orders of magnitude around physiologically realistic values. These extended expressions should enable finer characterization of the stochasticity of synaptic currents by analyzing experimentally recorded Vm distributions and may be applicable to other classes of stochastic processes as well.

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113. Synaptic and Temporal Ensemble Interpretation of Spike-Timing-Dependent Plasticity

Peter A. Appleby, Terry Elliott

https://doi.org/10.1162/0899766054796879 | pp. 2316-2336

We postulate that a simple, three-state synaptic switch governs changes in synaptic strength at individual synapses. Under this switch rule, we show that a variety of experimental results on timing-dependent plasticity can emerge from temporal and spatial averaging over multiple synapses and multiple spike pairings. In particular, we show that a critical window for the interaction of pre- and postsynaptic spikes

emerges as an ensemble property of the collective system, with individual synapses exhibiting only a minimal form of spike coincidence detection. In addition, we show that a Bienenstock-Cooper-Munro-like, rate-based plasticity rule emerges directly from such a model. This demonstrates that two apparently separate forms of neuronal plasticity can emerge from a much simpler rule governing the plasticity of individual synapses.

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114. What Can a Neuron Learn with Spike-Timing-Dependent Plasticity?

Robert Legenstein, Christian Naeger, Wolfgang Maass

https://doi.org/10.1162/0899766054796888 | pp. 2337-2382

Spiking neurons are very flexible computational modules, which can implement with different values of their adjustable synaptic parameters an enormous variety of different transformations F from input spike trains to output spike trains. We examine in this letter the question to what extent a spiking neuron with biologically realistic models for dynamic synapses can be taught via spike-timing-dependent plasticity (STDP) to implement a given transformation F. We consider a supervised learning paradigm where during training, the output of the neuron is clamped to the target signal (teacher forcing). The well-known perceptron convergence theorem asserts the convergence of a simple supervised learning algorithm for drastically simplified neuron models (McCulloch-Pitts neurons). We show that in contrast to the perceptron convergence theorem, no theoretical guarantee can be given for the convergence of STDP with teacher forcing that holds for arbitrary input spike patterns. On the other hand, we prove that average case versions of the perceptron convergence theorem hold for STDP in the case of uncorrelated and correlated Poisson input spike trains and simple models for spiking neurons. For a wide class of cross-correlation functions of the input spike trains, the resulting necessary and sufficient condition can be formulated in terms of linear separability, analogously as the well-known condition of learnability by perceptrons. However, the linear separability criterion has to be applied here to the columns of the correlation matrix of the Poisson input. We demonstrate through extensive computer simulations that the theoretically predicted convergence of STDP with teacher forcing also holds for more realistic models for neurons, dynamic synapses, and more general input distributions. In addition, we show through computer simulations that these positive learning results hold not only for the common interpretation of STDP, where STDP changes the weights of synapses, but also for a more realistic interpretation suggested by experimental data where STDP modulates the initial release probability of dynamic synapses.

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115. How Membrane Properties Shape the Discharge of Motoneurons: A Detailed Analytical Study

Claude Meunier, Karol Borejsza

https://doi.org/10.1162/0899766054796923 | pp. 2383-2420

Electrophysiological experiments and modeling studies have shown that afterhyperpolarization regulates the discharge of lumbar motoneurons in anesthetized cats and is an important determinant of their firing properties. However, it is still unclear how firing properties depend on slow afterhyperpolarization, input conductance, and the fast currents responsible for spike generation. We study a single-compartment integrate-and-fire model with a slow potassium conductance that exponentially decays between spikes. We show that this model is analytically solvable, and we investigate how passive and active membrane

properties control the discharge. We show that the model exhibits three distinct firing ranges (primary, secondary, and high frequency), and we explain the origin of these three ranges. Explicit expressions are established for the gain of the steady-state current-frequency (I-f) curve in the primary range and for the gain of the I-f curve for the first interspike interval. They show how the gain is controlled by the maximal conductance and the kinetic parameters of the afterhyperpolarization conductance. The gain also depends on the spike voltage threshold, and we compute how it is decreased by threshold accommodation (i.e., the increase of the threshold with the injected current). In contrast, the gain is not very sensitive to the input conductance. This implies that tonic synaptic activity shifts the current-frequency curve in its primary range, in agreement with experiments. Taking into account the absolute refractory period associated with spikes somewhat reduces the gain in the primary range. More importantly, it accounts for the approximately linear and steep secondary range observed in many motoneurons. In the nonphysiological high-frequency range, the behavior of the I-f curve is determined by the fast voltage-dependent currents, via the amplitude of the fast repolarization afterspike, the duration of the refractory period, and voltage threshold accommodation, if present.

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116. Stimulus Competition by Inhibitory Interference

Paul Tiesinga

https://doi.org/10.1162/0899766054796905 | pp. 2421-2453

When two stimuli are present in the receptive field of a V4 neuron, the firing rate response is between the weakest and strongest response elicited by each of the stimuli when presented alone (Reynolds, Chelazzi, & Desimone, 1999). When attention is directed toward the stimulus eliciting the strongest response (the preferred stimulus), the response to the pair is increased, whereas the response decreases when attention is directed to the other stimulus (the poor stimulus). When attention is directed to either of the two stimuli presented alone, the firing rate remains the same or increases slightly, but the coherence between the neuron's spike train and the local field potential can increase (Fries, Reynolds, Rorie, & Desimone, 2001). These experimental results were reproduced in a model of a V4 neuron under the assumption that attention modulates the activity of local interneuron networks. The V4 model neuron received stimulus-specific excitation from V2 and synchronous inhibitory inputs from two local interneuron networks in V4. Each interneuron network was driven by stimulus-specific excitatory inputs from V2 and was modulated by the activity of the frontal eye fields. Stimulus competition was present because of a delay in arrival time of synchronous volleys from each interneuron network. For small delays, the firing rate was close to the rate elicited by the preferred stimulus alone, whereas for larger delays, it approached the firing rate of the poor stimulus. When either stimulus was presented alone, the neuron's response was not altered by the change in delay, but could change due to modulation of the degree of synchrony of the corresponding interneuron network. The model suggests that top-down attention biases the competition between V2 columns for control of V4 neurons primarily by changing the relative timing of inhibition, whereas changes in the degree of synchrony of interneuron networks modulate the response to a single stimulus. The new mechanism proposed here for attentional modulation of firing rate, gain modulation by inhibitory interference, is likely to have more general applicability to cortical information processing.

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117. Optimization via Intermittency with a Self-Organizing Neural Network

Terence Kwok, Kate Smith■Miles
https://doi.org/10.1162/0899766054796860 | pp. 2454-2481

One of the major obstacles in using neural networks to solve combinatorial optimization problems is the convergence toward one of the many local minima instead of the global minima. In this letter, we propose a technique that enables a self-organizing neural network to escape from local minima by virtue of the intermittency phenomenon. It gives rise to novel search dynamics that allow the system to visit multiple global minima as meta-stable states. Numerical experiments performed suggest that the phenomenon is a combined effect of Kohonen-type competitive learning and the iterated softmax function operating near bifurcation. The resultant intermittent search exhibits fractal characteristics when the optimization performance is at its peak in the form of 1/f signals in the time evolution of the cost, as well as power law distributions in the meta-stable solution states. TheN-Queens problem is used as an example to illustrate the meta-stable convergence process that sequentially generates, in a single run, 92 solutions to the 8-Queens problem and 4024 solutions to the 17-Queens problem.

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118. Mixture Modeling with Pairwise, Instance-Level Class Constraints

Qi Zhao, David J. Miller
https://doi.org/10.1162/0899766054796914 | pp. 2482-2507

The goal of semisupervised clustering/mixture modeling is to learn the underlying groups comprising a given data set when there is also some form of instance-level supervision available, usually in the form of labels or pairwise sample constraints. Most prior work with constraints assumes the number of classes is known, with each learned cluster assumed to be a class and, hence, subject to the given class constraints. When the number of classes is unknown or when the one-cluster-per-class assumption is not valid, the use of constraints may actually be deleterious to learning the ground-truth data groups. We address this by (1) allowing allocation of multiple mixture components to individual classes and (2) estimating both the number of components and the number of classes. We also address new class discovery, with components void of constraints treated as putative unknown classes. For both real-world and synthetic data, our method is shown to accurately estimate the number of classes and to give favorable comparison with the recent approach of Shental, Bar-Hillel, Hertz, and Weinshall (2003).

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119. Geometrical Properties of Nu Support Vector Machines with Different Norms

Kazushi Ikeda, Noboru Murata
https://doi.org/10.1162/0899766054796897 | pp. 2508-2529

By employing the L1 or Linfinity norms in maximizing margins, support vector machines (SVMs) result in a linear programming problem that requires a lower computational load compared to SVMs with the L2 norm. However, how the change of norm affects the generalization ability of SVMs has not been clarified so far except for numerical experiments. In this letter, the geometrical meaning of SVMs with the Lp norm is investigated, and the SVM solutions are shown to have rather little dependency on p.

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120. The Effect of NMDA Receptors on Gain Modulation

Michiel Berends, Reinoud Maex, Erik De Schutter

https://doi.org/10.1162/089976605774320520 | pp. 2531-2547

The ability of individual neurons to modulate the gain of their input-output function is important for information processing in the brain. In a recent study, shunting inhibition was found to modulate the gain of cerebellar granule cells subjected to simulated currents through AMPA receptor synapses. Here we investigate the effect on gain modulation resulting from adding the currents mediated by NMDA receptors to a compartmental model of the granule cell. With only AMPA receptors, the changes in gain induced by shunting inhibition decreased gradually with the average firing rate of the afferent mossy fibers. With NMDA receptors present, this decrease was more rapid, therefore narrowing the bandwidth of mossy fiber firing rates available for gain modulation. The deterioration of gain modulation was accompanied by a reduced variability of the input current and saturation of NMDA receptors. However, when the output of the granule cell was plotted as a function of the average input current instead of the input firing frequency, both models showed very similar response curves and comparable gain modulation. We conclude that NMDA receptors do not directly impair gain control by shunting inhibition, but the effective bandwidth decreases as a consequence of the increased total charge transfer.

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121. Oscillatory Synchronization Requires Precise and Balanced Feedback Inhibition in a Model of the Insect Antennal Lobe

Dominique Martinez

https://doi.org/10.1162/089976605774320566 | pp. 2548-2570

In the insect olfactory system, odor-evoked transient synchronization of antennal lobe (AL) projection neurons (PNs) is phase-locked to the oscillations of the local field potential. Sensory information is contained in the spatiotemporal synchronization pattern formed by the identities of the phase-locked PNs. This article investigates the role of feedback inhibition from the local neurons (LNs) in this coding. First, experimental biological results are reproduced with a reduced computational spiking neural network model of the AL. Second, the low complexity of the model leads to a mathematical analysis from which a lower bound on the phase-locking probability is derived. Parameters involved in the bound indicate that PN phase locking depends not only on the number of LN-evoked inhibitory postsynaptic potentials (IPSPs) previously received, but also on their temporal jitter. If the inhibition received by a PN at the current oscillatory cycle is both perfectly balanced (i.e., equal to the mean inhibitory drive) and precise (without any jitter), then the PN will be phase-locked at the next oscillatory cycle with probability one.

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122. Response Properties of an Integrate-and-Fire Model That Receives Subthreshold Inputs

Xuedong Zhang, Laurel H. Carney

https://doi.org/10.1162/089976605774320584 | pp. 2571-2601

A computational technique is described for calculation of the interspike interval and poststimulus time histograms for the responses of an integrate-and-fire model to arbitrary inputs. The effects of the model parameters on the response statistics were studied systematically. Specifically, the probability distribution of

the membrane potential was calculated as a function of time, and the mean interspike interval and PST histogram were calculated for arbitrary inputs. For stationary inputs, the regularity of the output was studied in detail for various model parameters. For nonstationary inputs, the effects of the model parameters on the output synchronization index were explored. The results show that enhanced synchronization in response to low-frequency stimuli required a large number (n > 25) of weak inputs. Irregular responses and a linear input-output rate relationship required strong (but subthreshold) inputs with a small time constant. A model cell with mixed-amplitude synaptic inputs can respond to stationary inputs irregularly and have enhanced synchronization to nonstationary inputs that are phase-locked to low-frequency inputs. Both of these response properties have been reported for some cells in the ventral cochlear nucleus in the auditory brainstem.

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123. Incremental Online Learning in High Dimensions

Sethu Vijayakumar, Aaron D'Souza, Stefan Schaal

https://doi.org/10.1162/089976605774320557 | pp. 2602-2634

Locally weighted projection regression (LWPR) is a new algorithm for incremental nonlinear function approximation in high-dimensional spaces with redundant and irrelevant input dimensions. At its core, it employs nonparametric regression with locally linear models. In order to stay computationally efficient and numerically robust, each local model performs the regression analysis with a small number of univariate regressions in selected directions in input space in the spirit of partial least squares regression. We discuss when and how local learning techniques can successfully work in high-dimensional spaces and review the various techniques for local dimensionality reduction before finally deriving the LWPR algorithm. The properties of LWPR are that it (1) learns rapidly with second-order learning methods based on incremental training, (2) uses statistically sound stochastic leave-one-out cross validation for learning without the need to memorize training data, (3) adjusts its weighting kernels based on only local information in order to minimize the danger of negative interference of incremental learning, (4) has a computational complexity that is linear in the number of inputs, and (5) can deal with a large number of-possibly redundant-inputs, as shown in various empirical evaluations with up to 90 dimensional data sets. For a probabilistic interpretation, predictive variance and confidence intervals are derived. To our knowledge, LWPR is the first truly incremental spatially localized learning method that can successfully and efficiently operate in very high-dimensional spaces.

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124. On the Nonlearnability of a Single Spiking Neuron

Ji■■■ Š■■ma, Ji■■■ Sgall

https://doi.org/10.1162/089976605774320601 | pp. 2635-2647

We study the computational complexity of training a single spiking neuron N with binary coded inputs and output that, in addition to adaptive weights and a threshold, has adjustable synaptic delays. A synchronization technique is introduced so that the results concerning the nonlearnability of spiking neurons with binary delays are generalized to arbitrary real-valued delays. In particular, the consistency problem for N with programmable weights, a threshold, and delays, and its approximation version are proven to be NP-complete. It follows that the spiking neurons with arbitrary synaptic delays are not properly PAC learnable and do not allow robust learning unless RP = NP. In addition, the representation problem for N, a

question whether an n-variable Boolean function given in DNF (or as a disjunction of O(n) threshold gates) can be computed by a spiking neuron, is shown to be coNP-hard.

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125. A Novel Model-Based Hearing Compensation Design Using a Gradient-Free Optimization Method

Zhe Chen, Suzanna Becker, Jeff Bondy, Ian C. Bruce, S. Haykin

https://doi.org/10.1162/089976605774320575 | pp. 2648-2671

We propose a novel model-based hearing compensation strategy and gradient-free optimization procedure for a learning-based hearing aid design. Motivated by physiological data and normal and impaired auditory nerve models, a hearing compensation strategy is cast as a neural coding problem, and a Neurocompensator is designed to compensate for the hearing loss and enhance the speech. With the goal of learning the Neurocompensator parameters, we use a gradient-free optimization procedure, an improved version of the ALOPEX that we have developed, to learn the unknown parameters of the Neurocompensator. We present our methodology, learning procedure, and experimental results in detail; discussion is also given regarding the unsupervised learning and optimization methods.

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126. A Robust Information Clustering Algorithm

Qing Song

https://doi.org/10.1162/089976605774320548 | pp. 2672-2698

We focus on the scenario of robust information clustering (RIC) based on the minimax optimization of mutual information (MI). The minimization of MI leads to the standard mass-constrained deterministic annealing clustering, which is an empirical risk-minimization algorithm. The maximization of MI works out an upper bound of the empirical risk via the identification of outliers (noisy data points). Furthermore, we estimate the real risk VC-bound and determine an optimal cluster number of the RIC based on the structural risk-minimization principle. One of the main advantages of the minimax optimization of MI is that it is a nonparametric approach, which identifies the outliers through the robust density estimate and forms a simple data clustering algorithm based on the square error of the Euclidean distance.

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127. Learning Curves for Stochastic Gradient Descent in Linear Feedforward Networks

Justin Werfel, Xiaohui Xie, H. Sebastian Seung

https://doi.org/10.1162/089976605774320539 | pp. 2699-2718

Gradient-following learning methods can encounter problems of implementation in many applications, and stochastic variants are sometimes used to overcome these difficulties. We analyze three online training methods used with a linear perceptron: direct gradient descent, node perturbation, and weight perturbation. Learning speed is defined as the rate of exponential decay in the learning curves. When the scalar parameter that controls the size of weight updates is chosen to maximize learning speed, node perturbation is slower than direct gradient descent by a factor equal to the number of output units; weight perturbation is slower still by an additional factor equal to the number of input units. Parallel perturbation allows faster

learning than sequential perturbation, by a factor that does not depend on network size. We also characterize how uncertainty in quantities used in the stochastic updates affects the learning curves. This study suggests that in practice, weight perturbation may be slow for large networks, and node perturbation can have performance comparable to that of direct gradient descent when there are few output units. However, these statements depend on the specifics of the learning problem, such as the input distribution and the target function, and are not universally applicable.

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128. Information Geometry of Interspike Intervals in Spiking Neurons

Kazushi Ikeda

https://doi.org/10.1162/089976605774320593 | pp. 2719-2735

An information geometrical method is developed for characterizing or classifying neurons in cortical areas, whose spike rates fluctuate in time. Under the assumption that the interspike intervals of a spike sequence of a neuron obey a gamma process with a time-variant spike rate and a fixed shape parameter, we formulate the problem of characterization as a semiparametric statistical estimation, where the spike rate is a nuisance parameter. We derive optimal criteria from the information geometrical viewpoint when certain assumptions are added to the formulation, and we show that some existing measures, such as the coefficient of variation and the local variation, are expressed as estimators of certain functions under the same assumptions.

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129. Erratum

(authors unknown)

https://doi.org/10.1162/089976605774320511 | pp. 2736-2754

No abstract available.

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130. Erratum

(authors unknown)

https://doi.org/10.1162/neco.2005.17.12.2736 | pp. 2736-2736

No abstract available.

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131. Index

(authors unknown)

https://doi.org/10.1162/neco.2005.17.12.2737 | pp. 2737-2754

No abstract available.